Sinais e Sistemas para Engenharia

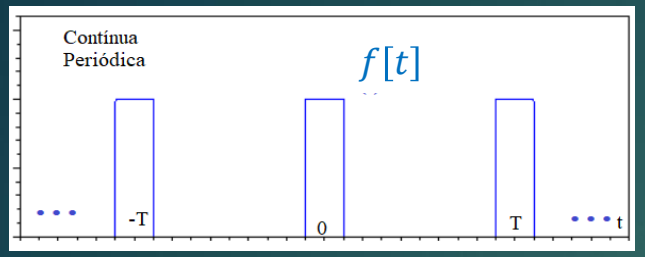
MÓDULO 14 - TRANSFORMADA Z DIRETA E INVERSA

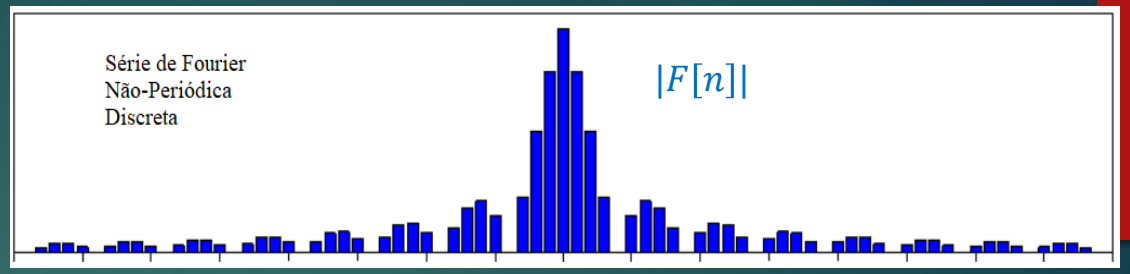
VERSÃO OCTAVE

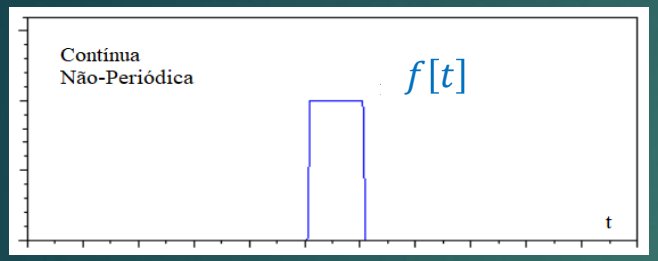
PROFESSOR LUCIANO NEVES DA FONSECA

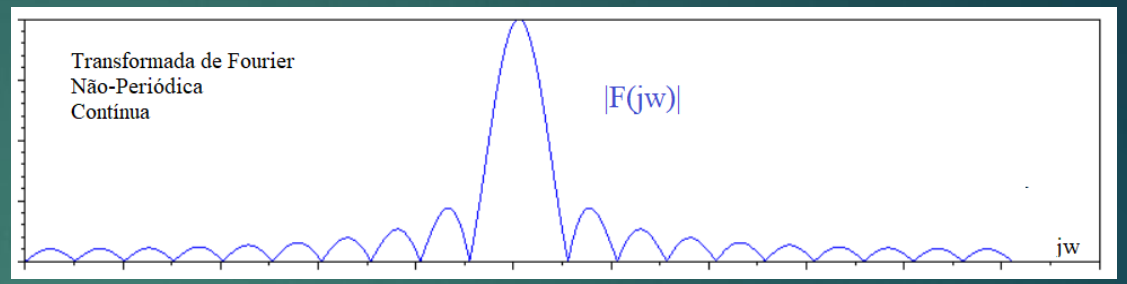
Transformada Z

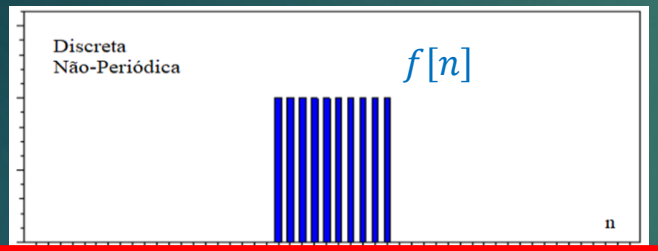
- A Transformada Z é a ferramenta de análise em frequência para o cálculo da resposta transiente e em regime permanente de sistemas discretos.
- A Transformada Z é então o equivalente discreto da Transformada de Laplace.
- A Transformada de Laplace é a ferramenta para se resolver equações diferenciais. A Transformada Z é a ferramenta para se resolver equações de diferenças.
- A Transformada Z é também uma generalização da Transformada de Fourier do Tempo discreto TFTD normalizada.

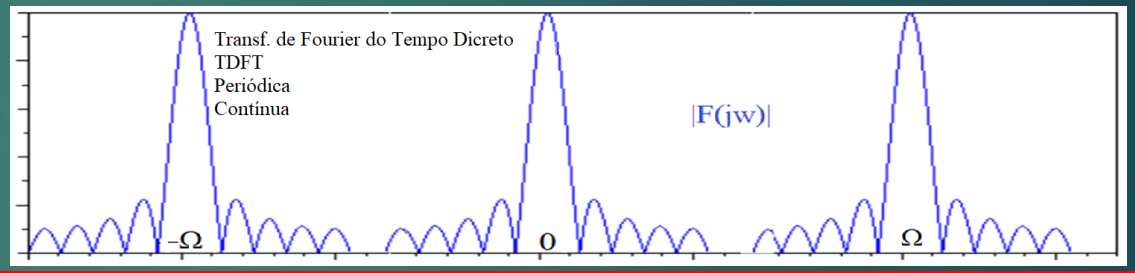

Contínua
Periódica
$f[t]$
-T
0
T
t

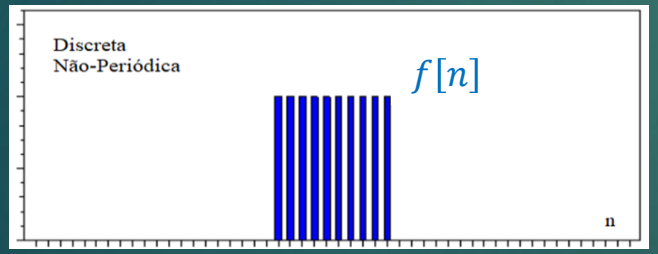

Série de Fourier
Não-Periódica
Discreta
$|F[n]|$

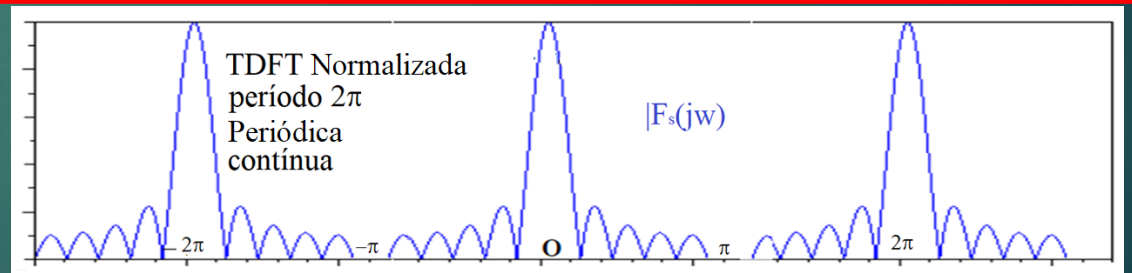

Contínua
Não-Periódica
$f[t]$
t

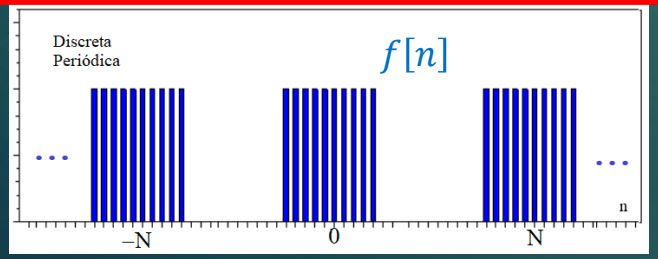

Transformada de Fourier
Não-Periódica
Contínua
|F(jw)|
jw

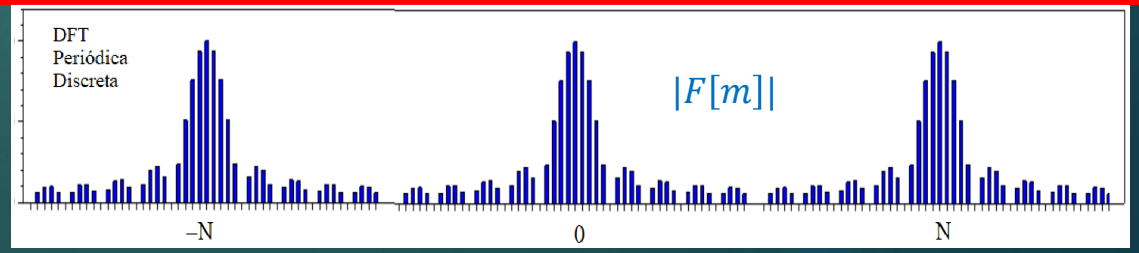

Discreta
Não-Periódica
$f[n]$
n


Transf. de Fourier do Tempo Dicreto
TDFT
Periódica
Contínua
|F(jw)|
−Ω
0
Ω


Discreta
Não-Periódica
$f[n]$
n


TDFT Normalizada
período 2π
Periódica
contínua
|Fs(jw)
−2π
−π
0
π
2π


Discreta
Periódica
$f[n]$
n
−N
0
N


DFT
Periódica
Discreta
$|F[m]|$
−N
0
N

A TFTD tem um relação direta com as transformadas de Laplace, de Fourier e Transformada Z

$$s = \sigma + jw$$

$$s = jw$$

$f(t)$

Laplace

$$F(s) = \int_0^{\infty} f(t)e^{-st}dt$$

$$s = jw$$

Fourier

$$F(jw) = \int_{-\infty}^{\infty} f(t)e^{-jwt}dt$$

$f(t)$

$$Amostragem \quad f_s = \frac{1}{T}$$

$$z = e^{sT}$$

$$Amostragem \; f_s = \frac{1}{T}$$

$$n = mT$$

$$z = e^{jw}$$

$f[n]$

Transformada Z

$$F_s(z) = \sum_{n=-\infty}^{\infty} f(n)\; z^{-n}$$

TFTD Normalizada

$$F_s(jw) = \sum_{n=-\infty}^{\infty} f(n)\; e^{-jwn}$$

$f[n]$

$$z = z_r + jz_i$$

$$s = jw$$

Definição

- A transformada Z descreve um sinal em termos de séries de potências da variável complexa exponencial 'z'.
- Se $y(n)$ for um sinal discreto, então $Y(z)$ é a Transformada Z do sinal $y(n)$

$$y[n] \xrightarrow{Z} Y(z) = \sum_{n=-\infty}^{\infty} y(n)z^{-n}$$

Relação entre z, s e w

- Por definição :

$$z = e^{s} = e^{\sigma + jw} = e^{\sigma}e^{jw} = re^{jw}; \quad com\ r = e^{\sigma}$$

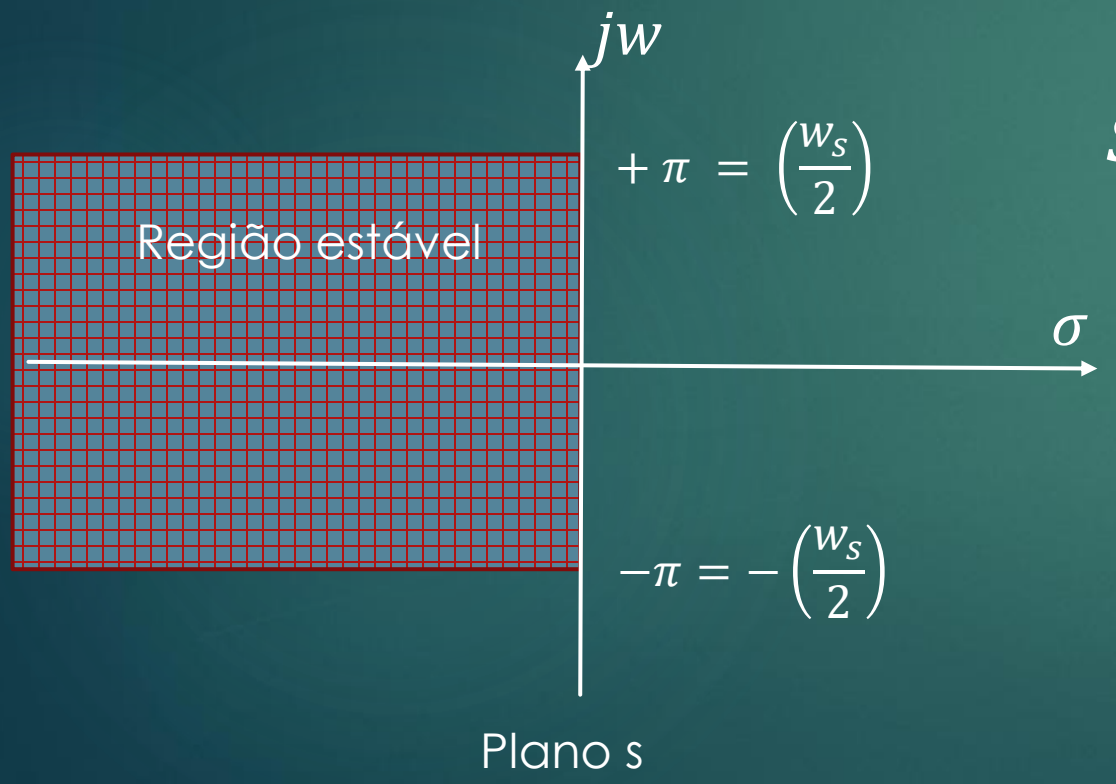
$$Y(z) = \sum_{-\infty}^{\infty} y(n)z^{-n} = \sum_{-\infty}^{\infty} y(n)\left(re^{jw}\right)^{-n} = \sum_{-\infty}^{\infty} y(n)r^{-n}e^{-jwn}$$

- Se $r = e^{\sigma}$=1, implica que σ=0, isto é , estamos no eixo imaginário do plano s
- Com r=1, a transformada Z se reduz à transformada de Fourier do Tempo Discreto TFTD.

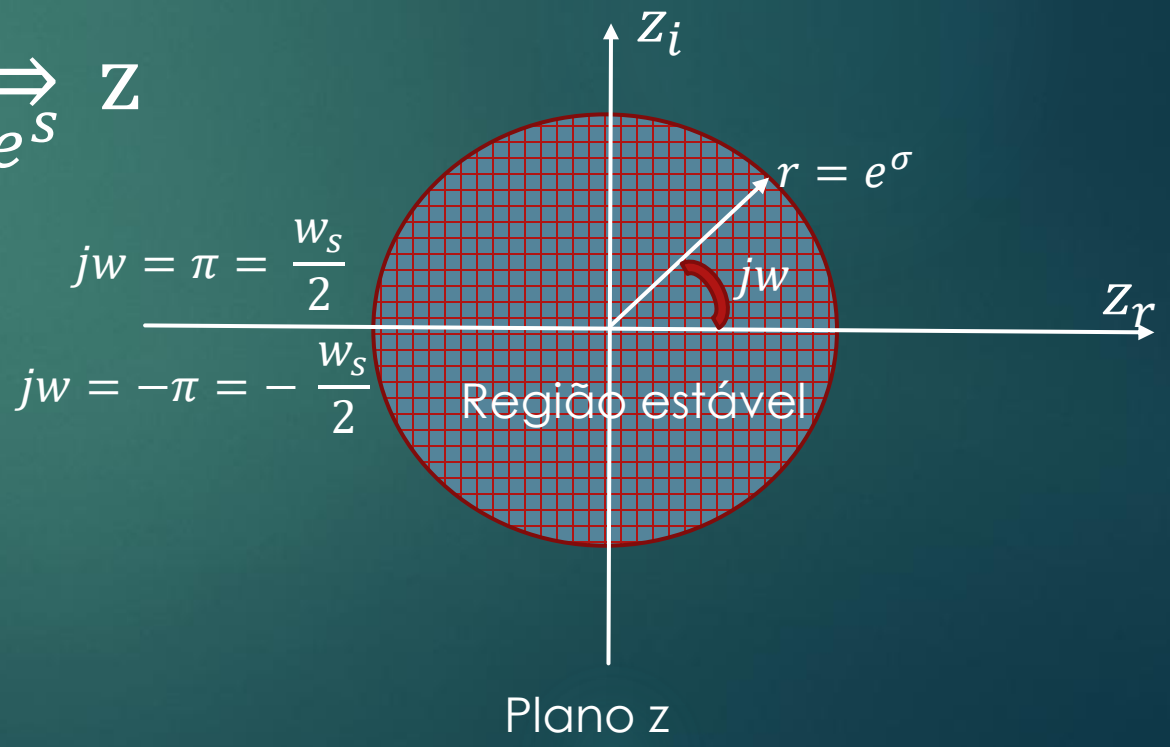
$$Y\left(z = e^{jw}\right) = \sum_{-\infty}^{\infty} y[n]e^{-jwn}$$

Mapeamento do Plano s para o Plano z

- O mapeamento do plano s para o plano Z se faz de uma reta vertical para um círculo $z = e^s$.
- Como vimos, a reta $\sigma = 0$ se mapeia no círculo unitário

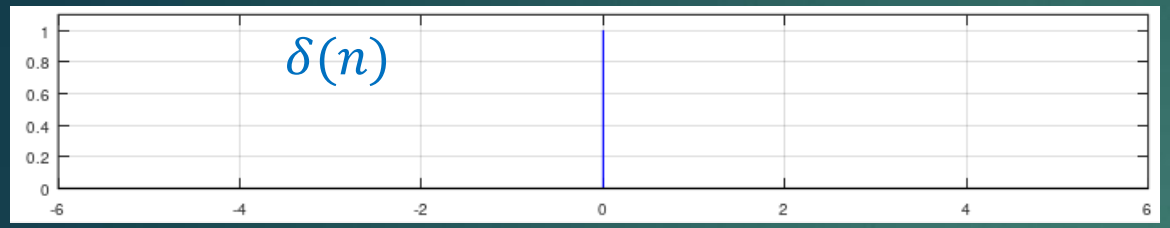


$$S \underset{z=e^s}{\Longleftrightarrow} Z$$



1 – Sinal delta $y[n] = \delta[n]$

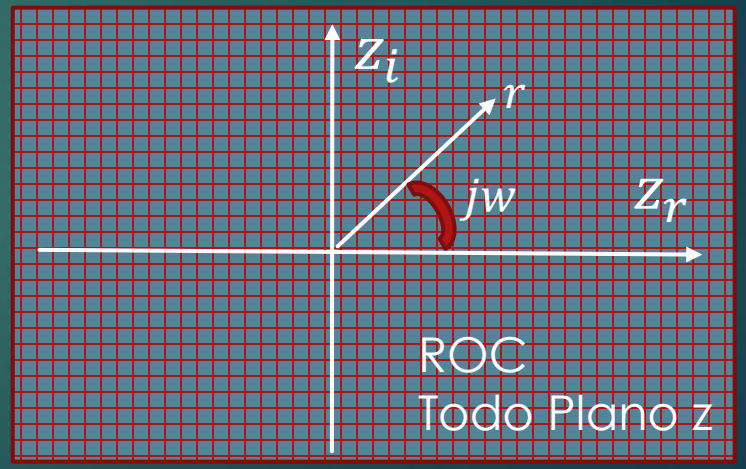
```
>> n=[-6:6];
>> stem(n,f_d(n),'b','marker','none');
```



$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n} = \sum_{n=0}^{0} \delta[n]\, z^{-n} = 1\, z^{0} = 1 \quad ; \quad \text{ROC} \quad \forall z$$

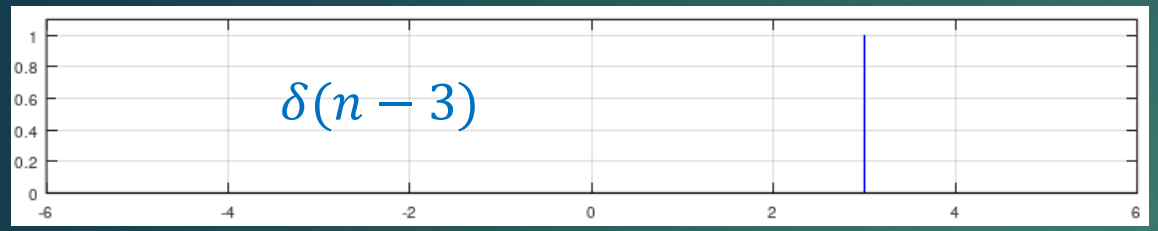
```
>> Yz=1;
```

```
>> Yz=tf(1)

Transfer function 'Yz' from input 'u1' to output ...

 y1:  1

Static gain.
```



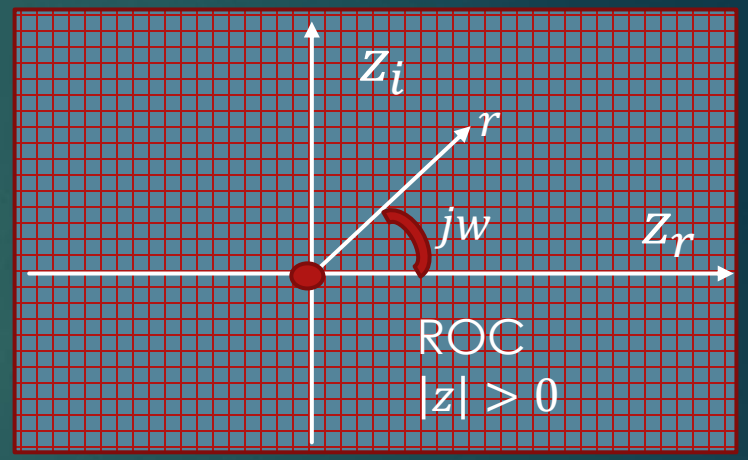
2 – Sinal delta com atraso $y[n] = \delta[n-k]$

```
>> n=[-6:6];
>> stem(n,f_d(n-3),'b','marker','none');
```



$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n} = \sum_{n=k}^{k} \delta[n-k]\, z^{-n} = 1\, \mathrm{z}^{-\mathrm{k}} = \mathrm{z}^{-\mathrm{k}} \quad ; \quad \mathrm{ROC} \;\; \forall\, z \mid \; |z| > 0$$

```
>> k=3;
>> z=tf('z',1);
>> Yz=z^-k

Transfer function 'Yz' from input 'u1' to output ...

        1
 y1:   ---
       z^3
```



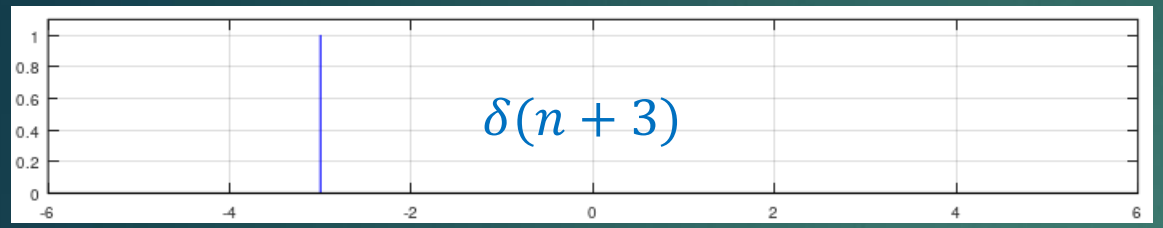
$$q = z^{-1}$$

$$Y(q) = \mathrm{q}^{\mathrm{k}}$$

```
>> k=3;
>> q=tf('q',1);
>> Yq=q^k

Transfer function 'Yq' from input 'u1' to output ...

 y1:  q^3
```

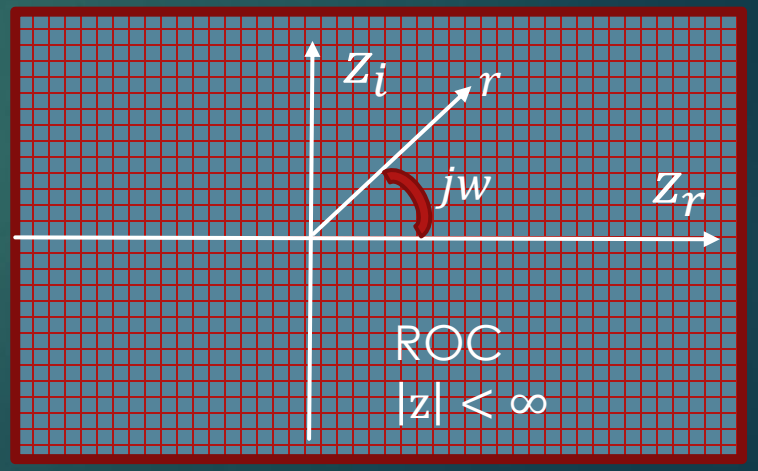
3 – Sinal delta adiantado $y[n] = \delta[n+k]$

```
>> n=[-6:6];
>> stem(n,f_d(n+3),'b','marker','none');
```



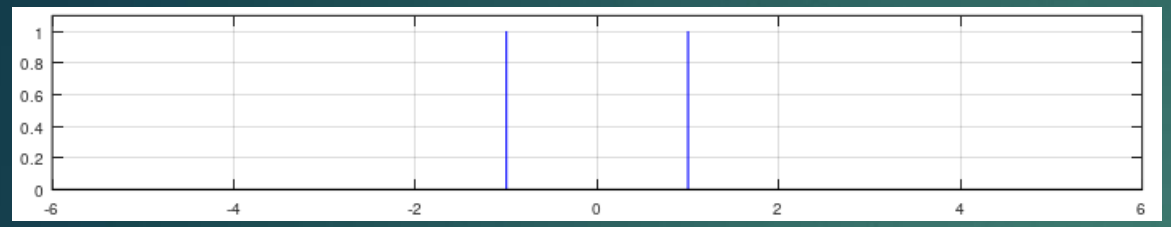
$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n} = \sum_{n=-k}^{-k} \delta[n+k]\, z^{-n} = 1\, \mathrm{z}^{-(-\mathrm{k})} = \mathrm{z}^{\mathrm{k}} \quad ; \quad \mathrm{ROC} \;\; \forall\, z \mid |z| < \infty$$

```
>> k=3;
>> z=tf('z',1);
>> Yz=z^k

Transfer function 'Yz' from input 'u1' to output ...

 y1:  z^3
```



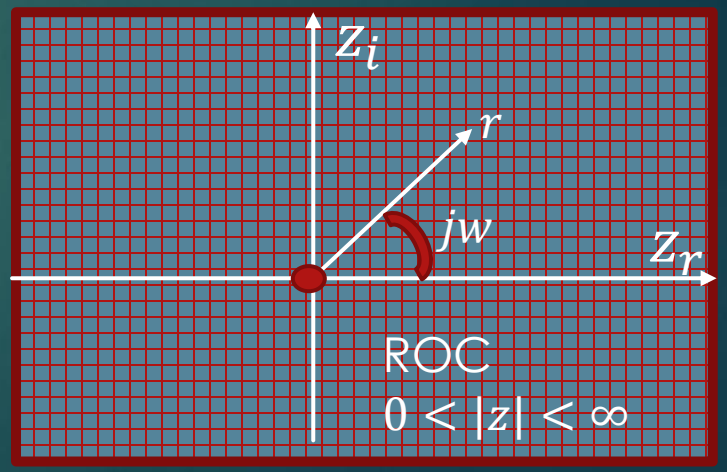
4 – 2 Sinais delta em k e -k $\quad y[n] = \delta[n+k] + \delta[n-k]$

```
>> n=[-6:6];
>> stem(n,f_d(n+1)+f_d(n-1),'b','marker','none');
```



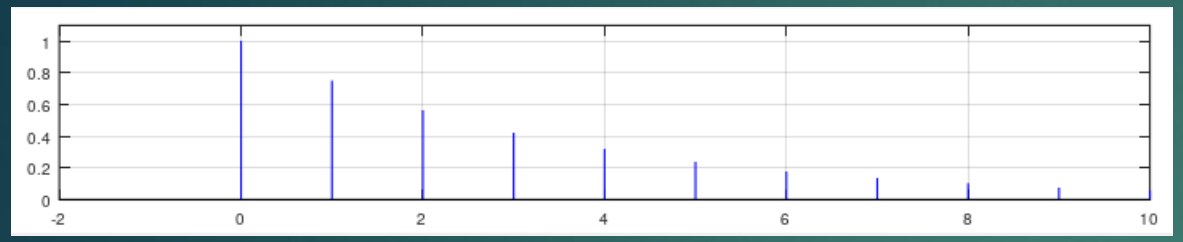
$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n} = \sum_{n=-\infty}^{\infty} \left[\delta[n+k] + \delta[n-k]\right] z^{-n} = \mathrm{z}^{\mathrm{k}} + \mathrm{z}^{-\mathrm{k}} \quad ; \quad \mathrm{ROC} \;\; \forall\, z \mid \;\; 0 < |z| < \infty$$

```
>> k=3;
>> Yz=z^k + z^-k

Transfer function 'Yz' from input 'u1' to output ...

      z^6 + 1
 y1:  -------
        z^3
```



5– Sinal exponencial causal $y[n] = \alpha^n;\ n \geq 0$

```
>> fn = @(n) (3/4).^n.*f_u(n);
>> n=[-2:10];
>> stem(n,fn(n),'b','marker','none');
```

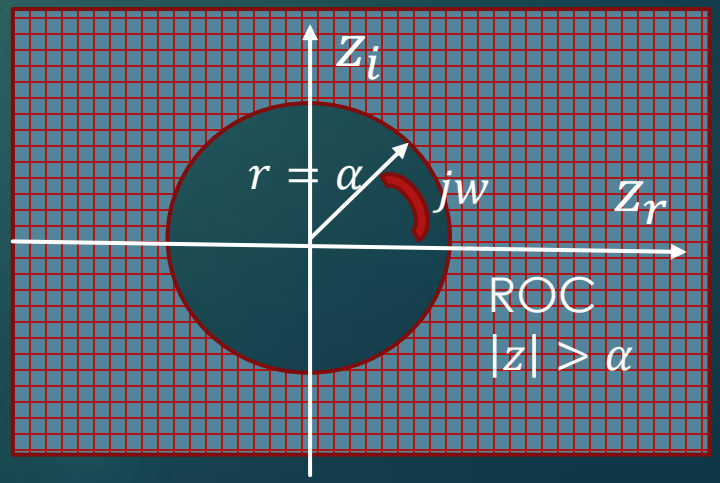


$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n} = \sum_{n=0}^{\infty} \alpha^n\, z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n = \frac{1}{1-\alpha z^{-1}} = \frac{z}{z-\alpha}; \quad \text{ROC} \quad \forall\, z \mid |z| > \alpha$$

```
>> Yz=tf([1,0],[1,-3/4],1)

Transfer function 'Yz' from input 'u1' to output ...

          z
 y1:  --------
      z - 0.75
```

```
>> a=3/4;
>> Yz=z/(z-a)

Transfer function 'Yz'

          z
 y1:  --------
      z - 0.75
```

Para haver convergência da somatória, a razão deve ser menor que 1: $|r| = |\alpha z^{-1}| < 1 \quad \rightarrow |z| > \alpha$



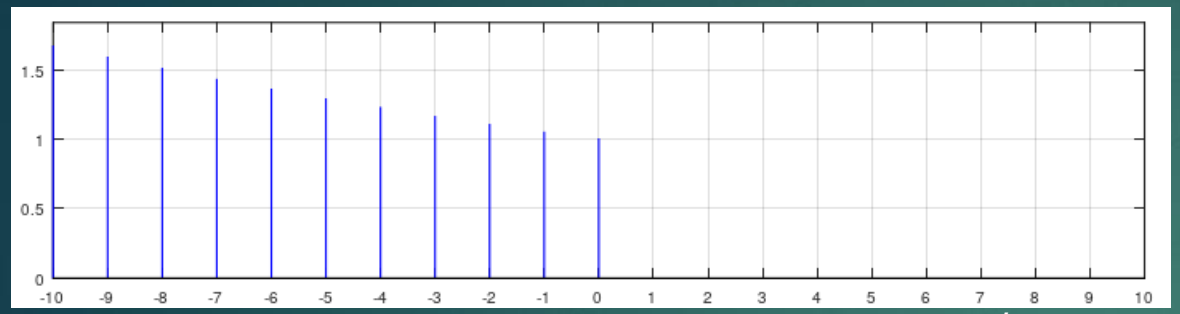
$$q = z^{-1}$$

$$Y(q) = \frac{1}{1-\alpha q}$$

```
>> Yq=tf(1,[-3/4 1],1,'Variable','q')

Transfer function 'Yq' from input 'u1' to output ...

           1
 y1:  -----------
      -0.75 q + 1
```

6) Sinal exponencial anti-causal $y[n] = \beta^n; \mathrm{n} \leq 0$

```
>> fn = @(n) (19/20).^n.*f_u(-n);
>> n=[-10:10];
>> stem(n,fn(n),'b','marker','none');
```



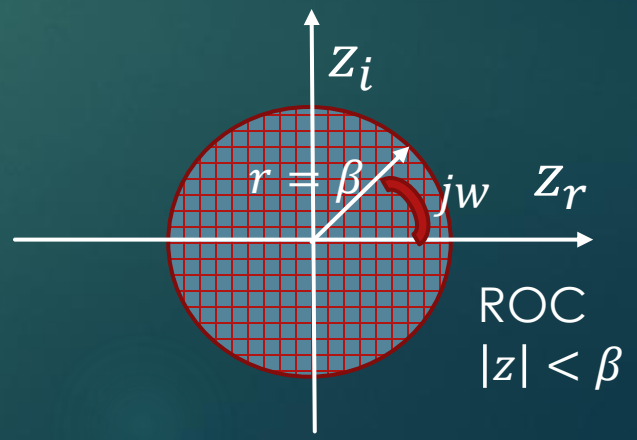
$$com\ m = -n, teremos:$$

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n} = \sum_{n=-\infty}^{0} \beta^n\, z^{-n} = \sum_{m=0}^{\infty} (\beta^{-1}z)^{-m} = \frac{1}{1-\beta^{-1}z} = \frac{\beta}{\beta - z}; \quad \mathrm{ROC} \quad \forall\, z \mid |z| < \beta$$

```
>> b=19/20;
>> Yz=b/(b-z)

Transfer function 'Yz' from input 'u1' to output ...

        0.95
 y1:  ---------
      -z + 0.95
```

Para haver convergência da somatória, a razão deve ser menor que 1:

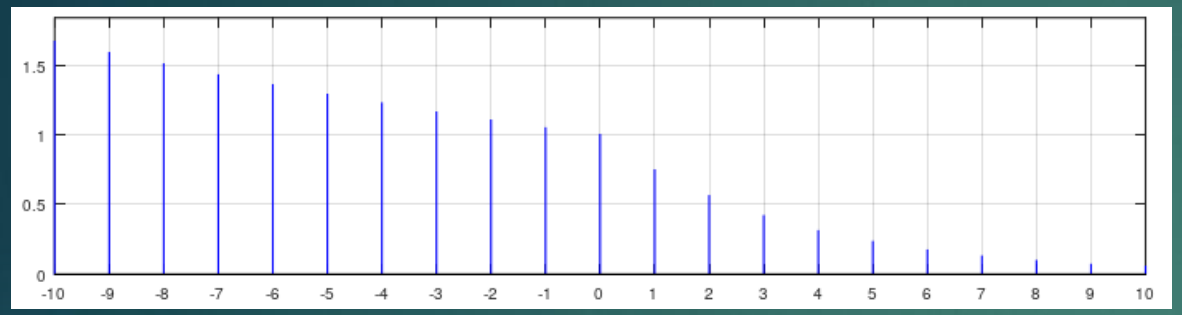
$$|r| = |\beta^{-1}z| < 1 \quad \rightarrow \quad |z| < \beta$$



7) Sinal exponencial causal + anti-causal

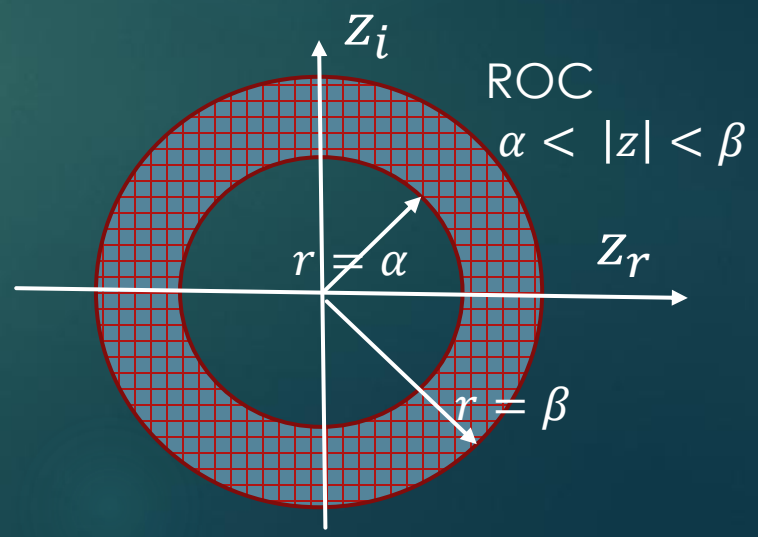
$$y[n] = \begin{cases} \alpha^n & ; n \geq 0 \\ \beta^n; & n < 0 \end{cases}$$

```
>> fn1 = @(n) (3/4).^n.*f_u(n);
>> fn2 = @(n) (19/20).^n.*f_u(-n-1);
>> n=[-10:10];
>> stem(n,fn1(n)+fn2(n),'b','marker','none');
```



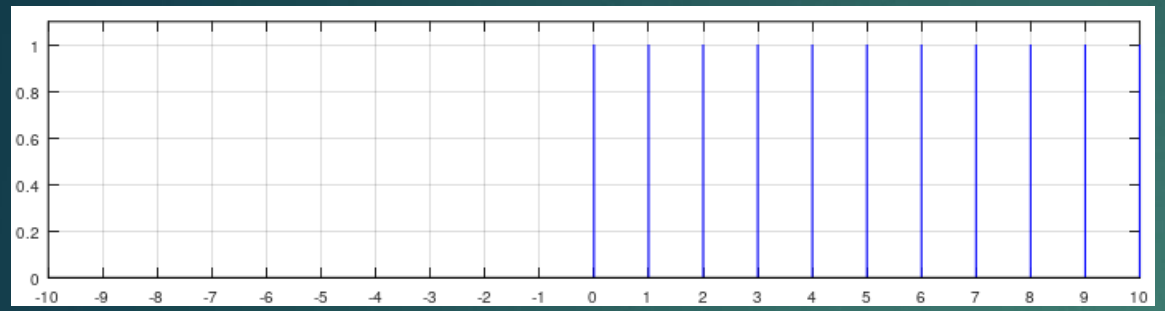
$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n + \sum_{m=1}^{\infty} (\beta z^{-1})^{-m} = \frac{z}{z-\alpha} + \frac{z}{\beta - z}; \quad \text{ROC} \quad \forall\, z \mid \quad \alpha < |z| < \beta$$

```
>> a=3/4;
>> b=19/20;
>> Yz=z/(z-a)+z/(b-z)

Transfer function 'Yz' from input 'u1' to output ...

                0.2 z
 y1:  ---------------------
      -z^2 + 1.7 z - 0.7125
```



8) Degrau unitário $y[n] = u[n]$

```
>> n=[-10:10];
>> stem(n,f_u(n),'b','marker','none');
```

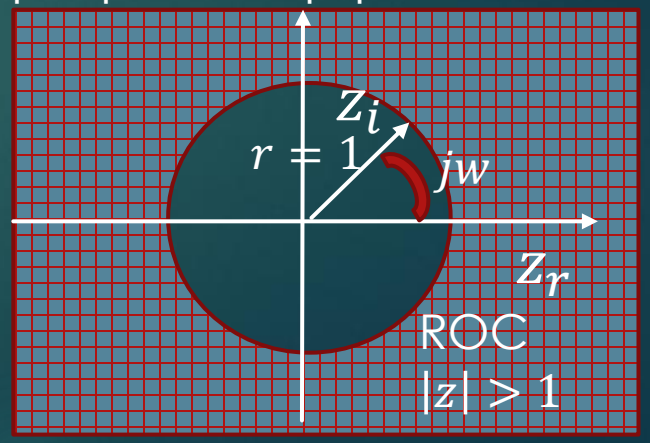


$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n} = \sum_{n=0}^{\infty} (1\, z^{-1})^n = \frac{1}{1 - z^{-1}} = \frac{z}{z-1}; \quad \text{ROC} \quad \forall\, z \mid \quad |z| > 1$$

```
>> Yz=z/(z-1)

Transfer function 'Yz' from input 'u1' to output ...

        z
 y1:  -----
      z - 1
```

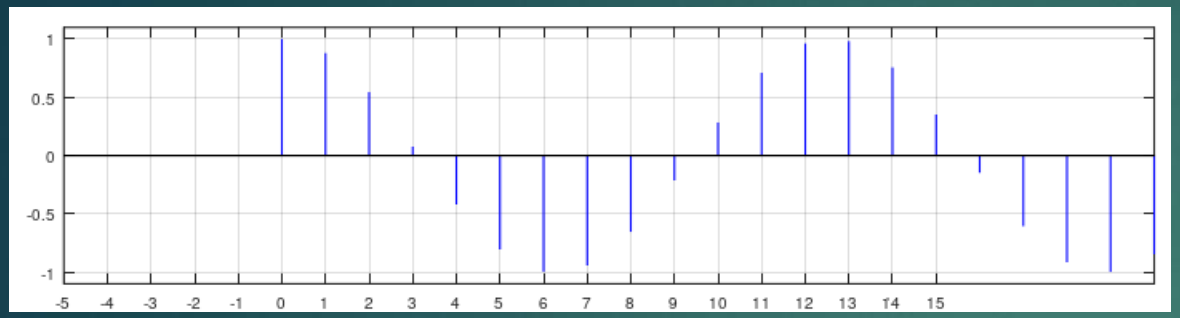
Para haver convergência da somatórias, as razão deve ser menores que 1:

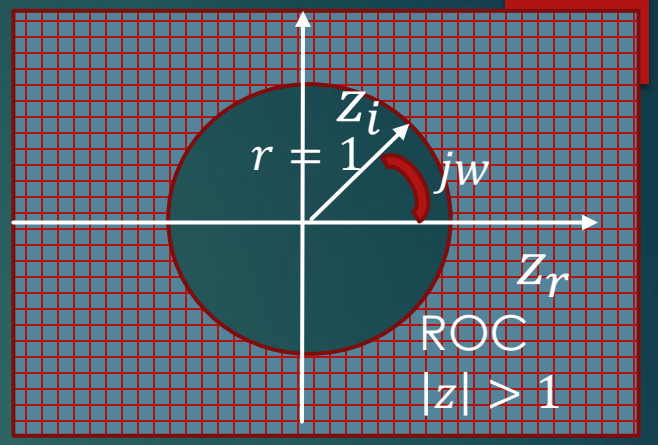
$$|r| = |z^{-1}| < 1 \quad \rightarrow |\, z| > 1$$



9) Cosseno $y[n] = cos(kn);\ \ n \geq 0$

```
>> fn = @(n) cos(0.5*n).*f_u(n);
>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none');
```





$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n} = \sum_{n=0}^{\infty} \cos(kn)\, z^{-n} = \sum_{n=0}^{\infty} \left(\frac{e^{jkn} + e^{-jkn}}{2}\right) z^{-n} = \frac{1}{2}\left(\frac{z}{z - e^{jk}} + \frac{z}{z - e^{-jk}}\right) = \frac{1}{2}\left(\frac{2z^2 - 2\cos(k)\, z}{z^2 - 2\cos(k)\, z + 1}\right)$$

Para haver convergência das somatórias, as razões devem ser menores que 1:

Então $|r| = \left|e^{jk} z^{-1}\right| < 1 \quad \rightarrow |z| > \left|e^{jk}\right| \rightarrow |z| > 1$

$e \quad |r| = \left|e^{-jk} z^{-1}\right| < 1 \quad \rightarrow |z| > \left|e^{-jk}\right| \rightarrow |z| > 1;$

```
>> k=0.5;
>> Yz=1/2*(2*z^2-2*cos(k)*z)/(z^2-2*cos(k)*z+1)

Transfer function 'Yz' from input 'u1' to output ...

       z^2 - 0.8776 z
 y1:  -----------------
      z^2 - 1.755 z + 1
```

```
>> k=0.5;
>> Yz=1/2*tf([2,-2*cos(k),0],[1,-2*cos(k),1],1)

Transfer function 'Yz' from input 'u1' to output ...

       z^2 - 0.8776 z
 y1:  -----------------
      z^2 - 1.755 z + 1
```

- Qual a resposta em frequência (Diagrama de Bode) de $y[n] = cos(kn);\ \ n \geq 0$

$$y[n] = cos(kn);\ n \geq 0 \xrightarrow{Z} Y(z) = \sum_{0}^{\infty} y[n]z^{-n} = \frac{1}{2}\left(\frac{2z^2 - 2\cos(k)\,z}{z^2 - 2\cos(k)\,z + 1}\right)$$

$$z = e^{jw}$$

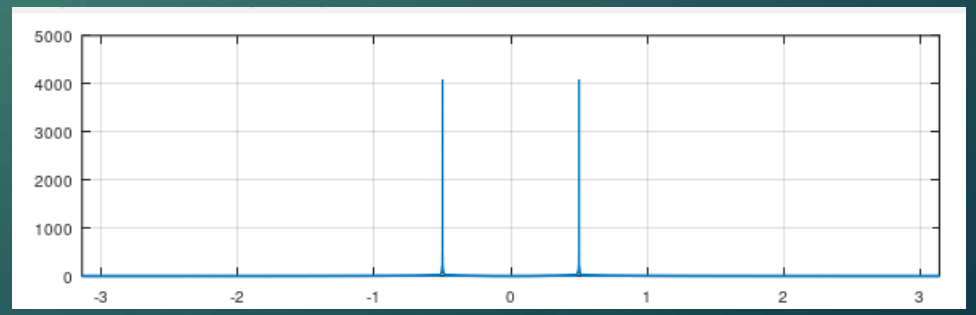
```
>> k=0.5;
>> Yz=1/2*tf([2,-2*cos(k),0],[1,-2*cos(k),1],1)

Transfer function 'Yz' from input 'u1' to output ...

       z^2 - 0.8776 z
 y1:  -----------------
      z^2 - 1.755 z + 1
```

$$Y(jw) = Y\left(z = e^{jw}\right) = \frac{1}{2}\left(\frac{2e^{j2w} - 2\cos(k)\,e^{jw}}{e^{2jw} - 2\cos(k)\,e^{jw} + 1}\right)$$

$$Y\left(z = e^{jw}\right)$$

```
>> w=linspace(-pi,pi,10000);
>> Yw=hval_fga(Yz,exp(i*w));
>> plot(w,abs(Yw))
```

- k é a frequência digital normalizada $2\pi\left(\frac{w}{w_s}\right)$
- π corresponde então à frequência $\frac{w_s}{2}$, que é a maior frequência normalizada possível.



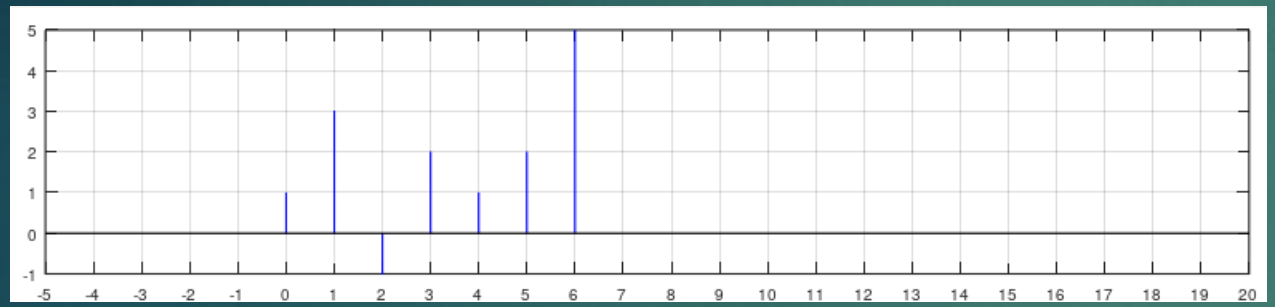
Resposta em Frequência k=05

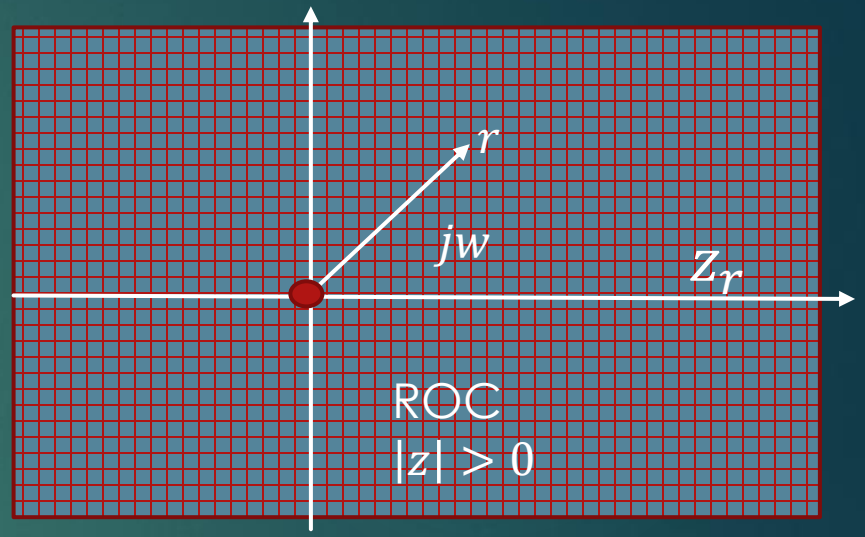
9) $y[n] = [1, 3, -1, 2, 1, 2, 5]$

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n} = (1)z^{0} + (3)z^{-1} + (-1)z^{-2} + (2)z^{-3} + (1)z^{-4} + (2)z^{-5} + 5z^{-6} \quad ; \quad \text{ROC} \quad \forall z \mid \quad |z| > 0$$

$$Y(z) = 1 + 3z^{-1} - z^{-2} + 2z^{-3} + z^{-4} + 2z^{-5} + 5z^{-6}$$

```
>> y=[1,3,-1,2,1,2,5];
>> N=length(y);
>> n=[0:N-1];
>> stem(n,y,'b','marker','none');
```





$$q = z^{-1}$$

$$Y(q) = 1 + 3q - q^{2} + 2q^{3} + q^{4} + 2q^{5} + 5q^{6}$$

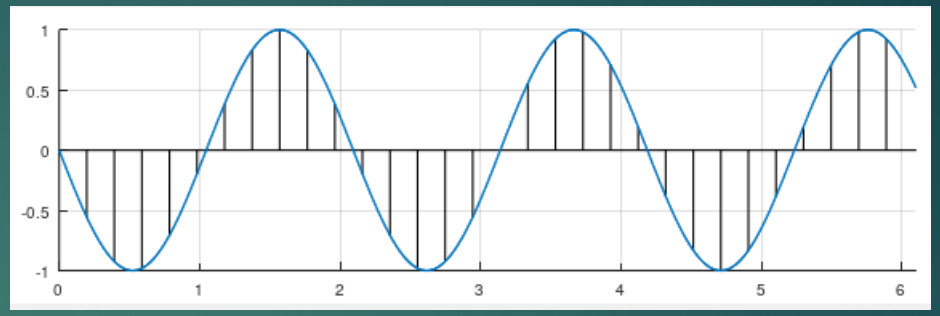
```
>> Yz=1+3*z^-1-z^-2+2*z^-3+z^-4+2*z^-5+5*z^-6;
```

```
>> Yz=tf(y,[1,zeros(1,N-1)],1)

Transfer function 'Yz' from input 'u1' to output ...

      z^6 + 3 z^5 - z^4 + 2 z^3 + z^2 + 2 z + 5
 y1:  ----------------------------------------
                          z^6

Sampling time: 1 s
Discrete-time model.
```

```
>> Yq=tf(flip(y),1,1,'Variable','q')

Transfer function 'Yq' from input 'u1' to output ...

 y1:  5 q^6 + 2 q^5 + q^4 + 2 q^3 - q^2 + 3 q + 1

Sampling time: 1 s
Discrete-time model.
```

9) $y[n]$- amostras do sinal $y(t) = -sen(3.0t)$, obtidas com $ws = 32$, durante T=6.1s

```
>> ft = @(t) -sin(3*t);
>> [ys,n]=sample_signal3(ft,32/(2*pi),0,6.1,true);
N=31 n=[0:30] fs=5.092958 Ts=0.196350 P=6.100000
```

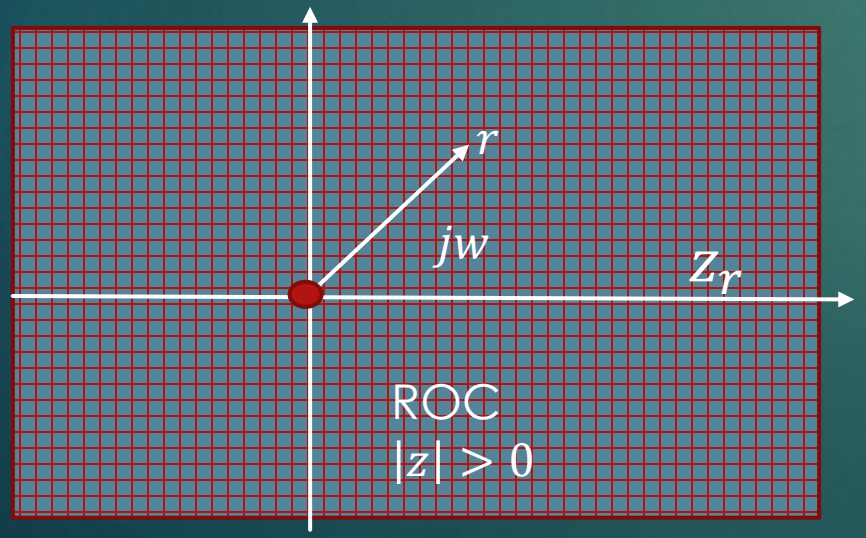


$y(n) =$

```
>> [n' ys']
ans =

         0         0
    1.0000   -0.5556
    2.0000   -0.9239
    3.0000   -0.9808
    4.0000   -0.7071
    5.0000   -0.1951
    6.0000    0.3827
    7.0000    0.8315
    8.0000    1.0000
    9.0000    0.8315
   10.0000    0.3827
   11.0000   -0.1951
   12.0000   -0.7071
   13.0000   -0.9808
   14.0000   -0.9239
   15.0000   -0.5556
   16.0000   -0.0000
   17.0000    0.5556
   18.0000    0.9239
   19.0000    0.9808
   20.0000    0.7071
   21.0000    0.1951
   22.0000   -0.3827
   23.0000   -0.8315
   24.0000   -1.0000
   25.0000   -0.8315
   26.0000   -0.3827
   27.0000    0.1951
   28.0000    0.7071
   29.0000    0.9808
   30.0000    0.9239
```

$$Y(z) = \sum_{n=-\infty}^{\infty} y[n]\, z^{-n}$$

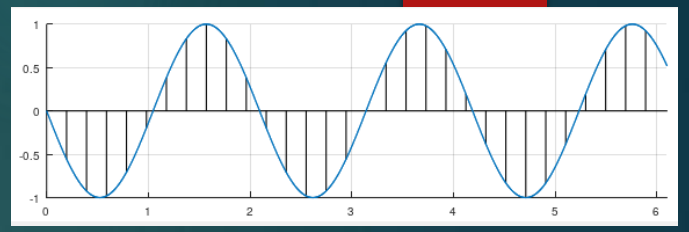
; ROC $\forall z \mid \quad |z| > 0$



- Qual a resposta em frequência (Diagrama de Bode) de F(z)

$$Y(jw) = Y(z = e^{jw})$$

```
>> ft = @(t) -sin(3*t);
>> [ys,n]=sample_signal3(ft,32/(2*pi),0,6.1,true);
N=31 n=[0:30] fs=5.092958 Ts=0.196350 P=6.100000
>> N=length(ys);
```




```
>> Yz=tf(ys,[1,zeros(1,N-1)],1)
>>
Transfer function 'Yz' from input 'u1' to output ...

      -0.5556 z^29 - 0.9239 z^28 - 0.9808 z^27 - 0.7071 z^26 - 0.1951 z^25 + 0.3827 z^24 + 0.8315 z^23 +
 z^22 + 0.8315 z^21 + 0.3827 z^20 - 0.1951 z^19 - 0.7071 z^18 - 0.9808 z^17 - 0.9239 z^16 - 0.5556 z^15
- 3.674e-16 z^14 + 0.5556 z^13 + 0.9239 z^12 + 0.9808 z^11 + 0.7071 z^10 + 0.1951 z^9 - 0.3827 z^8 - 0.8
315 z^7 - z^6 - 0.8315 z^5 - 0.3827 z^4 + 0.1951 z^3 + 0.7071 z^2 + 0.9808 z + 0.9239
 y1:  ----------------------------------------------------------------------------------------------------
----------------------------------------------------------------------------------------------------------
----------------------------------------------------------------------------------------------------------
-----------------------------------------------------------------------------------------

                                                                                                    z^30
```

$$f_d = \frac{f_{hertz}}{fs} = \frac{w_{rad}}{w_s} = \frac{3}{32}$$

$$w_n = 2\pi f_d = 2\pi \frac{3}{32} = 0.58$$

```
>> Yq=tf(flip(ys),1,1,'Variable','q')

Transfer function 'Yq' from input 'u1' to output ...

 y1:  0.9239 q^30 + 0.9808 q^29 + 0.7071 q^28 + 0.1951 q^27 - 0.3827 q^26 - 0.8315 q^25 - q^24 - 0.8315
q^23 - 0.3827 q^22 + 0.1951 q^21 + 0.7071 q^20 + 0.9808 q^19 + 0.9239 q^18 + 0.5556 q^17 - 3.674e-16 q^1
6 - 0.5556 q^15 - 0.9239 q^14 - 0.9808 q^13 - 0.7071 q^12 - 0.1951 q^11 + 0.3827 q^10 + 0.8315 q^9 + q^8
 + 0.8315 q^7 + 0.3827 q^6 - 0.1951 q^5 - 0.7071 q^4 - 0.9808 q^3 - 0.9239 q^2 - 0.5556 q
```

Resposta em Frequência de F(z)

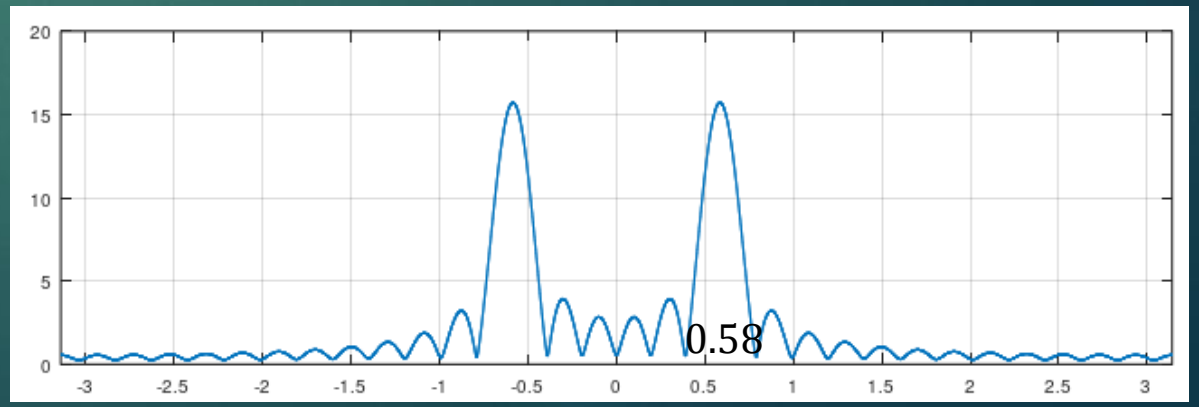
$$Y(z = e^{jw})$$

```
>> w=linspace(-pi,pi,10000);
>> Yw=hval_fga(Yz,exp(i*w));
>> plot(w,abs(Yw))
```

$$Y(q = e^{-jw})$$

```
>> Yw=hval_fga(Yq,exp(-i*w));
>> plot(w,abs(Yw));
```

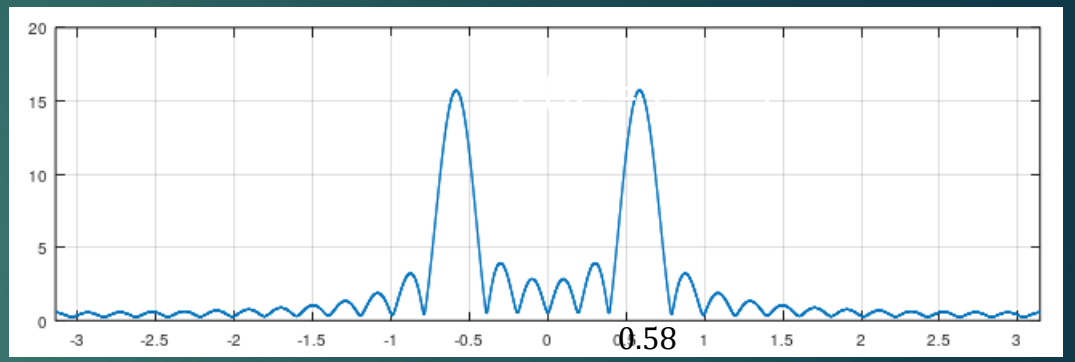
- $F_s(jw)$ é periódica com período 2π
- O eixo jw corresponde à frequência normalizada
- A frequência normalizada π corresponde à frequência real $\frac{w_s}{2}$



Resposta em frequência a partir de Y(z) ou Y(q)

```
>> Yq=tf(flip(ys),1,1,'Variable','q')

Transfer function 'Yq' from input 'u1' to output ...

 y1:  0.9239 q^30 + 0.9808 q^29 + 0.7071 q^28 + 0.1951 q^27 - 0.3827 q^26 - 0.8315 q^25 - q^24 - 0.8315
q^23 - 0.3827 q^22 + 0.1951 q^21 + 0.7071 q^20 + 0.9808 q^19 + 0.9239 q^18 + 0.5556 q^17 - 3.674e-16 q^1
6 - 0.5556 q^15 - 0.9239 q^14 - 0.9808 q^13 - 0.7071 q^12 - 0.1951 q^11 + 0.3827 q^10 + 0.8315 q^9 + q^8
 + 0.8315 q^7 + 0.3827 q^6 - 0.1951 q^5 - 0.7071 q^4 - 0.9808 q^3 - 0.9239 q^2 - 0.5556 q
```

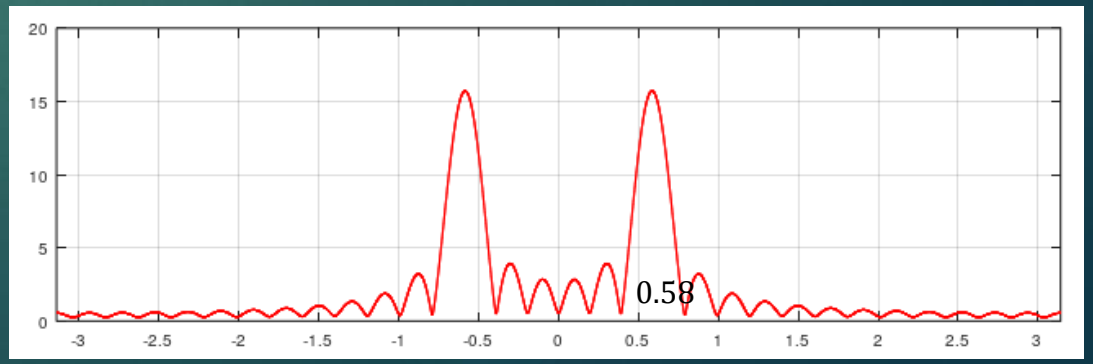
```
>> Yw=hval_fga(Yq,exp(-i*w));
>> plot(w,abs(Yw));
```



- Temos a opção de calcular a resposta em Frequência diretamente pelas amostras, utilizado para isso a definição da TDFT :

$$Y_s(jw) = \sum_{m=-\infty}^{\infty} y[n]\, e^{-jwn}$$

```
>> ft = @(t) -sin(3*t);
>> [ys,n]=sample_signal3(ft,32/(2*pi),0,6.1,false);
>> w=linspace(-pi,pi,10000);
>> Yw=tftd_fga(ys,w);
>> plot(w,abs(Yw),'r');
```



$f(n)$	$F(z) = \sum_{n=-\infty}^{\infty} f(n)\, z^{-n}$	**ROC**
$\delta[n]$	1	$plano\ Z$
$\delta[n-k]$	z^{-k}	$\lvert z\rvert > 0$
$\delta[n+k]$	z^{k}	$\lvert z\rvert < \infty$
$[a, b, c\ , d](0)$	$a + bz^{-1} + cz^{-2} + dz^{-3}$	$\lvert z\rvert > 0$
$u[n]$	$\frac{1}{1-z^{-1}} = \frac{z}{z-1}$	$\lvert z\rvert > 1$
$\alpha^n u[n]$	$\frac{1}{1-\alpha z^{-1}} = \frac{z}{z-\alpha}$	$\lvert z\rvert > \alpha$
$\beta^n u[-n]$	$\frac{\beta}{\beta - z}$	$\lvert z\rvert < \beta$
$\begin{cases} \alpha^n & ; n \geq 0 \\ \beta^n; & n < 0 \end{cases}$	$\frac{z}{z-\alpha} + \frac{z}{\beta - z}$	$\alpha < \lvert z\rvert < \beta$
$n\alpha^n u[n]$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\lvert z\rvert > \alpha$
$\cos(w_0 n)\, u[n]$	$\frac{(1 - z^{-1} cos(\mathrm{w_o}))}{1 - 2\cos(w_o)\, z^{-1} + z^{-2}}$	$\lvert z\rvert > 1$
$\mathrm{sen}(w_0 n)\, u[n]$	$\frac{z^{-1} sen(\mathrm{w_o})}{1 - 2\cos(w_o)\, z^{-1} + z^{-2}}$	$\lvert z\rvert > 1$
$\alpha^n \cos(w_0 n)\, u[n]$	$\frac{(1 - z^{-1}\alpha cos(\mathrm{w_o}))}{1 - 2\alpha\cos(w_o)\, z^{-1} + \alpha^2 z^{-2}}$	$\lvert z\rvert > \alpha$
$\alpha^n\, \mathrm{sen}(w_0 n)\, u[n]$	$\frac{z^{-1}\alpha sen(\mathrm{w_o})}{1 - 2\alpha\cos(w_o)\, z^{-1} + \alpha^2 z^{-2}}$	$\lvert z\rvert > \alpha$

$x(n)$	$X(z)=\sum_{n=0}^{\infty} x(n)z^{-n}$	ROC
$\delta(n)$	1	$\lvert z\rvert > 0$
$au(n)$	$\frac{az}{z-1}$	$\lvert z\rvert > 1$
$nu(n)$	$\frac{z}{(z-1)^2}$	$\lvert z\rvert > 1$
$n^2u(n)$	$\frac{z(z+1)}{(z-1)^3}$	$\lvert z\rvert > 1$
$a^n u(n)$	$\frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$
$e^{-na}u(n)$	$\frac{z}{(z-e^{-a})}$	$\lvert z\rvert > e^{-a}$
$na^n u(n)$	$\frac{az}{(z-a)^2}$	$\lvert z\rvert > \lvert a\rvert$
$\sin(an)u(n)$	$\frac{z\sin(a)}{z^2-2z\cos(a)+1}$	$\lvert z\rvert > 1$
$\cos(an)u(n)$	$\frac{z[z-\cos(a)]}{z^2-2z\cos(a)+1}$	$\lvert z\rvert > 1$
$a^n\sin(bn)u(n)$	$\frac{[a\sin(b)]z}{z^2-[2a\cos(b)]z+a^2}$	$\lvert z\rvert > \lvert a\rvert$
$a^n\cos(bn)u(n)$	$\frac{z[z-a\cos(b)]}{z^2-[2a\cos(b)]z+a^2}$	$\lvert z\rvert > \lvert a\rvert$
$e^{-an}\sin(bn)u(n)$	$\frac{[e^{-a}\sin(b)]z}{z^2-[2e^{-a}\cos(b)]z+e^{-2a}}$	$\lvert z\rvert > e^{-a}$
$e^{-an}\cos(bn)u(n)$	$\frac{z[z-e^{-a}\cos(b)]}{z^2-[2e^{-a}\cos(b)]z+e^{-2a}}$	$\lvert z\rvert > e^{-a}$
$2\lvert A\rvert\lvert P\rvert^n\cos(n\theta+\varphi)u(n)$ $P=\lvert P\rvert\angle\theta,\ A=\lvert A\rvert\angle\varphi$	$\frac{Az}{z-P}+\frac{A^*z}{z-P^*}$	

x(n) .	X(z)	ROC
$\delta[n]$	1	$\forall z$
$u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert < 1$
$\delta[n-m]$	z^{-m}	$0<\lvert z\rvert<\infty$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert < \lvert a\rvert$
$\cos(\omega_0 n)u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
$\sin(\omega_0 n)u[n]$	$\frac{\sin(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
$r^n\cos(\omega_0 n)u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
$r^n\sin(\omega_0 n)u[n]$	$\frac{r\sin(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
$\begin{cases} a^n, & 0\le n\le N-1, \\ 0, & \text{caso contrário} \end{cases}$	$\frac{1-a^Nz^{-N}}{1-az^{-1}}$	$\lvert z\rvert > 0$

Propriedades da Transformada Z

1) Linearidade

Sejam os sinais discretos f[n] e g[n],

Se $y[n] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{4}{3}\right)^n u[-n-1] \xrightarrow{Z} Y(z) = -\frac{z}{\left(z-\frac{1}{2}\right)\left(z-\frac{4}{3}\right)}$ com ROC $\frac{1}{2} < |z| < \frac{4}{3}$

e $g[n] = \left(\frac{1}{5}\right)^n u[n] - \left(\frac{1}{3}\right)^n u[n] \xrightarrow{Z} G(z) = \frac{z}{z-\frac{1}{5}} - \frac{z}{z-\frac{1}{3}}$ $com\ ROC$ R_g $|z| > \frac{1}{3}$

Então $ay[n] + bg[n] \xrightarrow{Z} aY(z) + bG(z)$ $Com\ ROC$ $R_f \cap R_g$ $com\ ROC$ $\frac{1}{3} < |z| < \frac{4}{3}$

2) Convolução no tempo

Sejam os sinais discretos y[n] e g[n]

Se $y[n] \xrightarrow{Z} Y(z)$ com ROC R_f

e $g[n] \xrightarrow{Z} G(z)$ $com\ ROC$ R_h

Então $y[n] * g[n] \xrightarrow{Z} Y(z)G(z)$ $com\ ROC$ $R_f \cap R_h$

Propriedades da Transformada Z

3) Deslocamento no tempo com condições iniciais nulas

Seja um sinal discreto $y[n]$

Se $y[n] \xrightarrow{Z} Y(z)$ $\quad com \quad ROC \quad R_f$

Então $y[n-k] \xrightarrow{z} Y(z)z^{-k}$ $\quad com \quad ROC \quad R_f \quad \cup \quad |z| > 0$

- Notar que um deslocamento no tempo corresponde à multiplicação por z^{-1} no domínio Z.

4) Deslocamento no tempo com condições iniciais

Seja um sinal discreto $y[n]$

Se $y[n] \xrightarrow{Z} Y(z)$ $\quad com \quad ROC \quad R_f$

Então $y[n-1] \xrightarrow{z} Y(z)z^{-1} + y[-1]$ $\quad com \quad ROC \quad R_f \quad \cup \quad |z| > 0$

Então $y[n-2] \xrightarrow{z} Y(z)z^{-2} + y[-1]z^{-1} + y[-2]$ $\quad com \quad ROC \quad R_f \quad \cup \quad |z| > 0$

Então $y[n-k] \xrightarrow{z} Y(z)z^{-k} + y[-1:-1:-k]z^{(k-1:0)}$ $\quad com \quad ROC \quad R_f \quad \cup \quad |z| > 0$

Propriedades da Transformada Z

5) Primeira Diferença no Tempo

Seja um sinal discreto $y[n]$

Se $y[n] \xrightarrow{z} Y(z)$ $com \quad ROC \quad R_f$

Então $y[n] - y[n-1] \xrightarrow{z} (1 - z^{-1})Y(z)$ $com \quad ROC \quad R_f$

Prova:

$$Z[y[n] - y[n-1]] = \sum_{n=-\infty}^{\infty} (y[n] - y[n-1])\, z^{-n} = Y(z) - z^{-1}Y(z) = (1 - z^{-1})Y(z)$$

6) Acumulação no Tempo

Seja um sinal discreto $y[n]$

Se $y[n] \xrightarrow{z} Y(z)$ $com \quad ROC \quad R_f$

Então $\sum_{k=-\infty}^{n} y[k] \xrightarrow{z} \frac{1}{(1-z^{-1})} Y(z)$ $com \quad ROC \quad R_f \cap \ (|z| > 1)$

Prova:

$$\sum_{k=-\infty}^{n} y[k] = u[n] * y[n] \xrightarrow{z} U(z)\, Yz = \frac{1}{(1 - z^{-1})} Y(z)$$

Propriedades da Transformada Z

7) Multiplicação por exponencial - Modulação

Seja os sinal discreto $y[n]$.

Se $y[n] \xrightarrow{Z} Y(z)$ $\quad com \quad ROC \quad R_f$

Então $a^n y[n] \xrightarrow{Z} Y\left(\frac{z}{a}\right)$ $\quad com \quad ROC \quad |a|R_f$

$$Y_1(z) = \sum_{n=-\infty}^{\infty} a^n y[n] z^{-n} = \sum_{n=-\infty}^{\infty} y[n] \left(\frac{z}{a}\right)^{-n} = Y\left(\frac{z}{a}\right)$$

$como\ z = e^{jw} \quad e \quad a = e^{jw_o}, \qquad então \quad Y\left(\frac{z}{a}\right) = Y\left(e^{j(w-wo)}\right) \quad - Modulação$

8) Expansão no tempo - escala

Seja um sinal discreto y(n)

Se $y[n] \xrightarrow{Z} Y(z)$ $\quad com \quad ROC \quad R_f$

Então $y\left[\frac{n}{k}\right] \xrightarrow{Z} Y(z^k)$ $\quad com \quad ROC \quad R_f^{\frac{1}{k}}$

Prova:

$$Z\left[y\left[\frac{n}{k}\right]\right] = \sum_{n=-\infty}^{\infty} y\left[\frac{n}{k}\right] z^{-n} = \sum_{m=-\infty}^{\infty} y[m] z^{-km} = Y(z^k)$$

Com n é inteiro, k não pode assumir qualquer valor, isto é, n deve ser múltiplo de k

$$y\left[\frac{n}{k}\right] = \begin{cases} y\left[\frac{n}{k}\right] se\ n\ é\ , múltiplo\ de\ k \\ 0 \quad caso\ contrário \end{cases}$$

Propriedades da Transformada Z

9) Diferenciação em Z

Seja um sinal discreto $y[n]$

Se $y[n] \xrightarrow{z} Y(z)$ com ROC R_f

Então $ny[n] \xrightarrow{z} -zY'(z)$ com ROC R_f

$$Y'(z) = \frac{d}{dz}\left[\sum_{n=-\infty}^{\infty} y[n]z^{-n}\right] = \sum_{n=-\infty}^{\infty} (-n)y(n)z^{-n-1} = -z^{-1}\sum_{n=-\infty}^{\infty} ny(n)z^{-n}$$

$$Z[ny[n]] = \sum_{n=-\infty}^{\infty} ny[n]z^{-n} = -zY'(z)$$

10) Inversão no tempo

Seja um sinal discreto $y[n]$

Se $y[n] \xrightarrow{z} Y(z)$ com ROC R_f

Então $y[-n] \xrightarrow{z} Y(z^{-1})$ com ROC $1/R_f$

- Notar que uma reflexão no tempo corresponde a substituir z por z^{-1} no domínio Z.'

Propriedades da Transformada Z

11) Teorema do valor inicial

Seja um sinal discreto $y[n]$

Se $y[n] \xrightarrow{z} Y(z)$ $\quad com \quad ROC \quad R_f$

Então $y[0] \xrightarrow{z} \lim_{z \to \infty} Y(z)$

12) Teorema do valor final

Seja um sinal discreto $y[n]$

Se $y[n] \xrightarrow{z} Y(z)$ $\quad com \quad ROC \quad R_f$

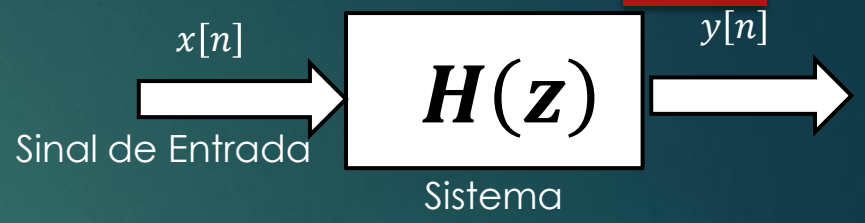
Então $\lim_{n \to \infty} y[n] \xrightarrow{z} \lim_{z \to 1} (z-1) Y(z)$

Transformada Z de uma equação de diferenças

$$se \quad x[n] \xRightarrow{\mathrm{Z}} X(z)$$

$$e \quad y[n] \xRightarrow{\mathrm{Z}} Y(z)$$

$$x[n] = 0 \; para \; n \leq 0$$
$$y[n] = 0 \; para \; n \leq 0$$



$$y[n] = \sum_{k=1}^{N=3} a_k y[n-k] + \sum_{k=0}^{M=2} b_k x[n-k] \xRightarrow{Z} Y(z) = a_1 z^{-1} Y(z) + a_2 z^{-2} Y(z) + a_3 z^{-3} Y(z) + b_0 X(z) + b_1 z^{-1} X(z) + b_2 z^{-2} X(z)$$

$$y[n] = a_1\, y[n-1] + a_2 y[n-2] + a_3 y[n-3] + b_0 x[n] + b_1 x[n-1] + b_2 x[n-2]$$

$$\mathrm{Z} \Downarrow$$

$$Y(z) = a_1 z^{-1} Y(z) + a_2 z^{-2} Y(z) + a_3 z^{-3} Y(z) + b_0 X(z) + b_1 z^{-1} X(z) + b_2 z^{-2} X(z)$$

Função de Transferência em Z

$$Y(z)(1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3}) = X(z)(b_0 + b_1 z^{-1} + b_2 z^{-2})$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{(1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3})}$$

Definição	$x(n)$	$\boldsymbol{X}(z) = \sum_{n=-\infty}^{\infty} f(n)\, z^{-n}$
Inversão no Tempo	$x[-n]$	$X(z^{-1})$
Linearidade	$ax[n] + by[n]$	$aX(z) + bY(z)$
Multiplicação por exponencial	$a^n x[n]$	$X\left(\frac{z}{a}\right)$
Atraso de n amostras	$x[n-m]$	$z^{-m}X(z)$
Adiantar n amostras	$x[n+m]$	$z^{m}X(z)$
Escala em frequência	$b^n x[n]$	$X\left(\frac{z}{b}\right)$
Diferenciação	$nx[n]$	$-zX'(z)$
Integração	$\frac{1}{n}x(n)$	$-\int \frac{X(z)}{z} dz$
Acumulação no Tempo Convolução $y(n) = 1$	$\sum_{i=0}^{n} x[n]$	$\frac{z}{z-1}X(z)$
Valor inicial	$x[0]$	$\lim_{z \to \infty} X(z)$
Valor Final	$x[\infty]$	$\lim_{z \to 1}(z-1)X(z)$
Primeira Diferença no Tempo	$x[n] - x[n-1]$	$\frac{1}{(1-z^{-1})}X(z)$
Expansão no tempo - escala	$x\left[\frac{n}{k}\right]$	$X(z^k)$
Convolução no tempo	$f[n] * g[n]$	$F(z)H(z)$

Métodos para inversão da Transformada Z

- 1) Inspeção Direta
- 2) Frações Parciais
- 3) Divisão Polinomial
- 4) Integral de Contorno

Inspeção direta

A ideia da inversão da transformada Z por inspeção direta se baseia no nosso conhecimento e familiaridade com a transformada Z direta. Reconhecemos que uma certa função X(z) é a transformada Z de um função x(n) no tempo discreto. Normalmente utilizamos tabelas para auxiliar na inversão por inspeção direta

$x(n)$	$X(z)=\sum_{n=0}^{\infty} x(n)z^{-n}$	ROC
$\delta(n)$	1	$\lvert z\rvert > 0$
$au(n)$	$\frac{az}{z-1}$	$\lvert z\rvert > 1$
$nu(n)$	$\frac{z}{(z-1)^2}$	$\lvert z\rvert > 1$
$n^2u(n)$	$\frac{z(z+1)}{(z-1)^3}$	$\lvert z\rvert > 1$
$a^n u(n)$	$\frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$
$e^{-na}u(n)$	$\frac{z}{(z-e^{-a})}$	$\lvert z\rvert > e^{-a}$
$na^n u(n)$	$\frac{az}{(z-a)^2}$	$\lvert z\rvert > \lvert a\rvert$
$\sin(an)u(n)$	$\frac{z\sin(a)}{z^2-2z\cos(a)+1}$	$\lvert z\rvert > 1$
$\cos(an)u(n)$	$\frac{z[z-\cos(a)]}{z^2-2z\cos(a)+1}$	$\lvert z\rvert > 1$
$a^n\sin(bn)u(n)$	$\frac{[a\sin(b)]z}{z^2-[2a\cos(b)]z+a^2}$	$\lvert z\rvert > \lvert a\rvert$
$a^n\cos(bn)u(n)$	$\frac{z[z-a\cos(b)]}{z^2-[2a\cos(b)]z+a^2}$	$\lvert z\rvert > \lvert a\rvert$
$e^{-an}\sin(bn)u(n)$	$\frac{[e^{-a}\sin(b)]z}{z^2-[2e^{-a}\cos(b)]z+e^{-2a}}$	$\lvert z\rvert > e^{-a}$
$e^{-an}\cos(bn)u(n)$	$\frac{z[z-e^{-a}\cos(b)]}{z^2-[2e^{-a}\cos(b)]z+e^{-2a}}$	$\lvert z\rvert > e^{-a}$
$2\lvert A\rvert\lvert P\rvert^n\cos(n\theta+\varphi)u(n)$ $P=\lvert P\rvert\angle\theta,\ A=\lvert A\rvert\angle\varphi$	$\frac{Az}{z-P}+\frac{A^*z}{z-P^*}$	

x(n)	X(z)	ROC
$\delta[n]$	1	$\forall z$
$u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert < 1$
$\delta[n-m]$	z^{-m}	$0<\lvert z\rvert<\infty$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert < \lvert a\rvert$
$\cos(\omega_0 n)u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
$\sin(\omega_0 n)u[n]$	$\frac{\sin(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
$r^n\cos(\omega_0 n)u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
$r^n\sin(\omega_0 n)u[n]$	$\frac{r\sin(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
$\begin{cases} a^n, & 0\le n\le N-1, \\ 0, & \text{caso contrário} \end{cases}$	$\frac{1-a^Nz^{-N}}{1-az^{-1}}$	$\lvert z\rvert > 0$

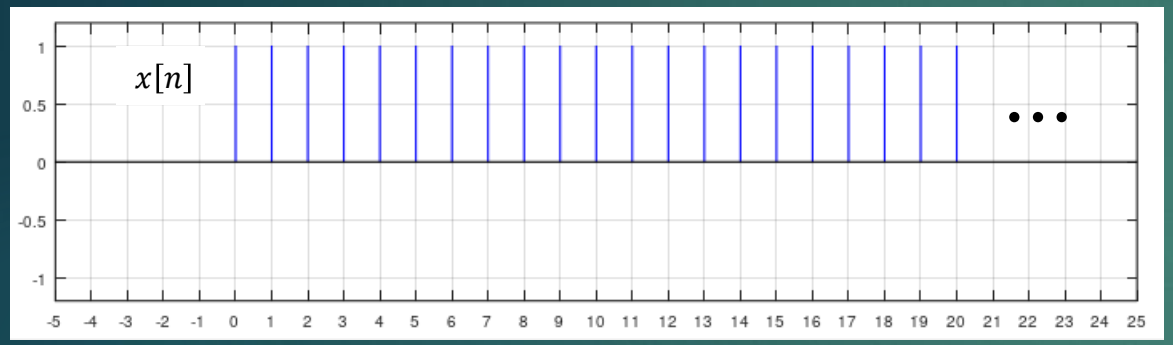
Calcular a transformadas Z inversa dos seguintes sinais no domínio Z

1) $X(z) = \frac{1}{1-z^{-1}} = \frac{z}{z-1}$

$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$a^n u(n)$	$\frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$

Causal

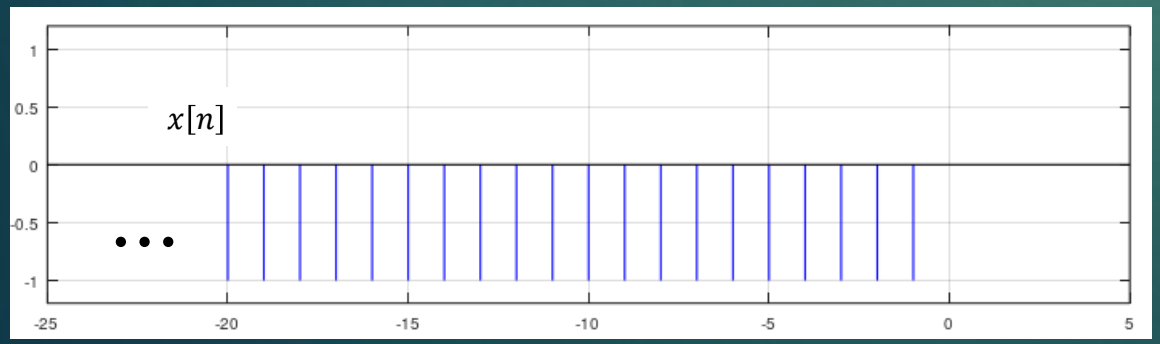
$$x[n] = (1)^n u[n] = \mathrm{u}[n]; \quad |\mathrm{z}| > 1$$




```
>> z=tf('z',1);
>> Xz=z/(z-1);
```

```
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+(1.00000 * f_u (n)))

>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none')
```

$$x[n] = -u[-n-1]; \quad |\mathrm{z}| < 1$$



$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{z}{z-a}$	$\lvert z\rvert < \lvert a\rvert$

Anticausal

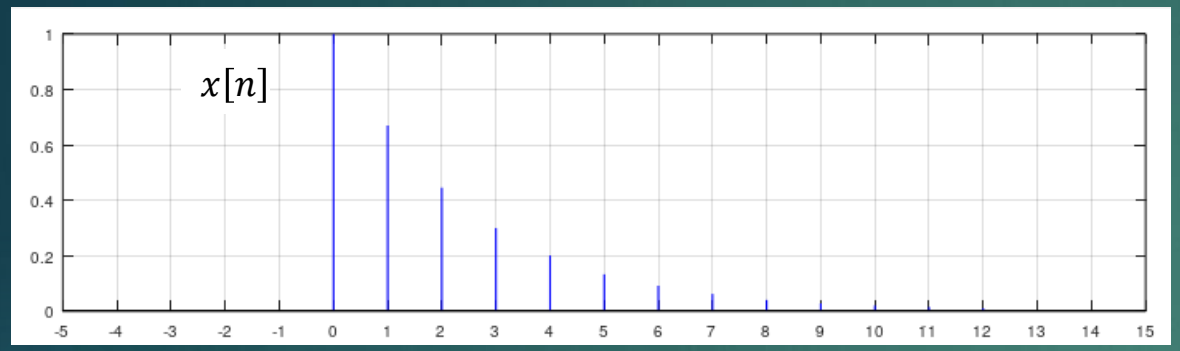
```
>> n=[-20:5];
>> stem(n,-f_u(-n-1),'b','marker','none');
```

2) $X(z) = \frac{1}{1-\frac{2}{3}z^{-1}} = \frac{1}{1-\frac{2}{3}q} = \frac{z}{z-\frac{2}{3}}$

$a^n u(n)$	$\frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$

```
>> Xz=1/(1-2/3*z^-1);
>> fn= itransformadaZ_Hz(Xz,1)
fn =

@(n) (+(1.00000 * (0.66667) .^ n .* f_u (n)))
```

$$x[n] = \left(\frac{2}{3}\right)^n u[n]; \quad |z| > a$$



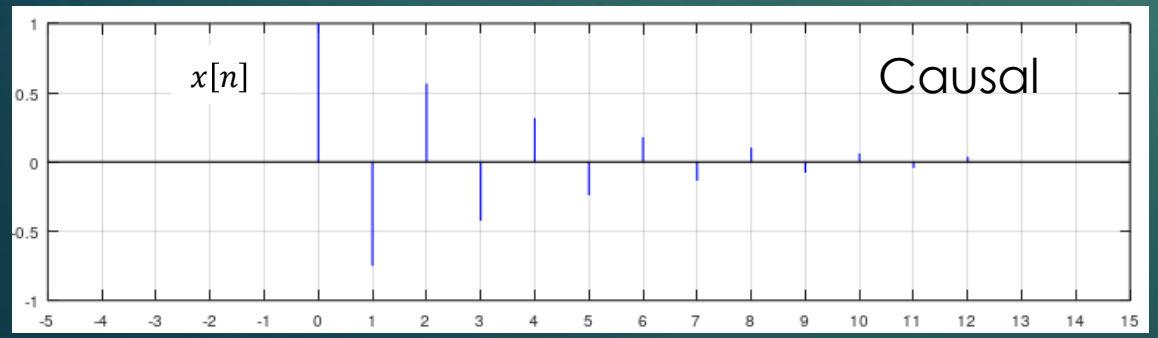

```
>> Xz=z/(z-2/3);
>> fn= itransformadaZ_Hz(Xz,1)
fn =

@(n) (+(1.00000 * (0.66667) .^ n .* f_u (n)))

>> n=[-5:12];
>> stem(n,fn(n),'b','marker','none');
```

3) $X(z) = \frac{z}{z+\frac{3}{4}} = \frac{1}{1+\frac{3}{4}z^{-1}} = \frac{1}{1+\frac{3}{4}q}$

$a^n u(n)$	$\frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$

```
>> Xz=1/(1+3/4*z^-1);
>> fn= itransformadaZ_Hz(Xz,1)
fn =

@(n) (+(1.00000 * (-0.75000) .^ n .* f_u (n)))
```

$$x[n] = \left(-\frac{3}{4}\right)^n u[n]; \quad |z| > a$$

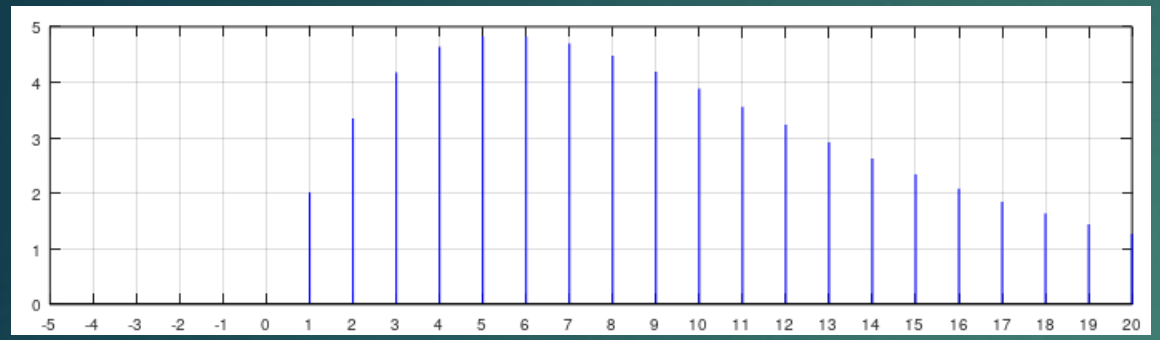



```
>> Xz=z/(z+3/4);
>> fn= itransformadaZ_Hz(Xz,1)
fn =

@(n) (+(1.00000 * (-0.75000) .^ n .* f_u (n)))

>> n=[-5:12];
>> stem(n,fn(n),'b','marker','none');
```

4) $X(z) = \frac{2z^{-1}}{\left(1-\frac{5}{6}z^{-1}\right)^2} = \frac{12}{5}\frac{\frac{5}{6}z^{-1}}{\left(1-\frac{5}{6}z^{-1}\right)^2}$

$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$

$$x[n] = \frac{12}{5} n \left(\frac{5}{6}\right)^n u[n]; \quad |z| > a$$



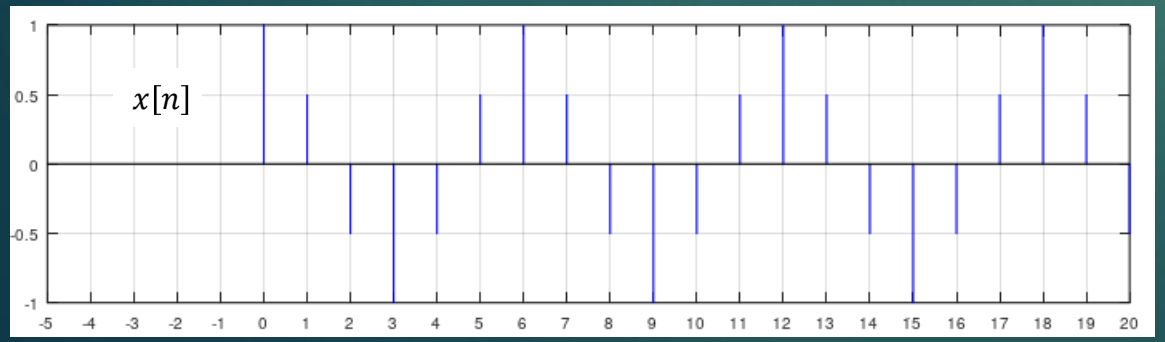
```
>> Xz=2*z^-1/(1-5/6*z^-1)^2;
>> fn= itransformadaZ_Hz(Xz,1)
fn =

@(n) (+(2.40000 * n .* (0.83333) .^ n .* f_u (n)))

>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none');
```

5) $X(z) = \frac{z^2-\frac{1}{2}z}{z^2-z+1} = \frac{z\left[z-\left(\frac{1}{2}\right)\right]}{z^2-2\left(\frac{1}{2}\right)z+1}$

$$a = \arccos\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$$x[n] = \cos\left(\frac{\pi}{3}n\right) u[n]; \quad |z| > 1$$

$\cos(an)u(n)$	$\frac{z[z-\cos(a)]}{z^2-2z\cos(a)+1}$	$\lvert z\rvert > 1$



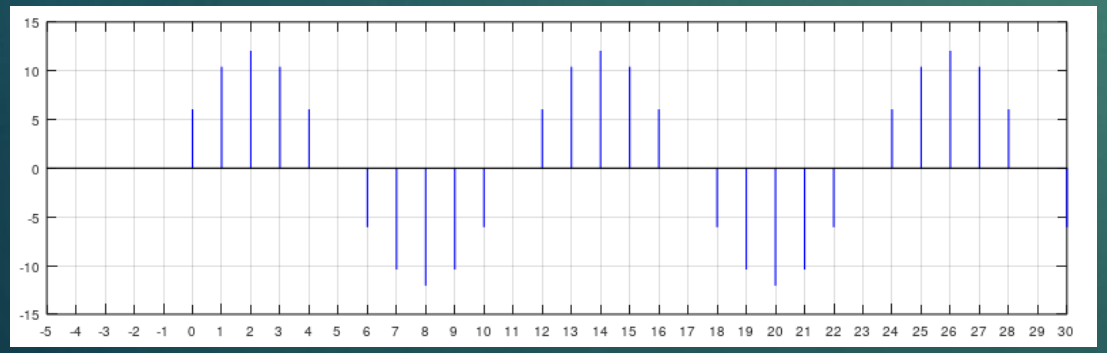

```
>> Xz=(z^2-1/2*z)/(z^2-z+1);
>> fn= itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+(2 * (0.50000) * cos (n * (1.04720) - (0.00000)) .* f_u (n)))

>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none');
```

6) $X(z) = \frac{6z^2}{z^2-\sqrt{3}z+1} = 6\left(\frac{z^2-z\frac{\sqrt{3}}{2}+z\frac{\sqrt{3}}{2}}{z^2-2\left(\frac{\sqrt{3}}{2}\right)z+1}\right) = 6\left(\frac{z\left(z-\frac{\sqrt{3}}{2}\right)}{z^2-2\left(\frac{\sqrt{3}}{2}\right)z+1}\right) + 6\sqrt{3}\left(\frac{z\frac{1}{2}}{z^2-2\left(\frac{\sqrt{3}}{2}\right)z+1}\right)$

$$x[n] = 12\left(\frac{1}{2}\cos\left(\frac{\pi}{6}n\right) + \frac{\sqrt{3}}{2}\sin\left(\frac{\pi}{6}n\right)\right)u[n]; \quad |z| > 1$$

$$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$
$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$\sin(an)u(n)$	$\frac{z\sin(a)}{z^2 - 2z\cos(a) + 1}$	$\lvert z\rvert > 1$
$\cos(an)u(n)$	$\frac{z[z - \cos(a)]}{z^2 - 2z\cos(a) + 1}$	$\lvert z\rvert > 1$



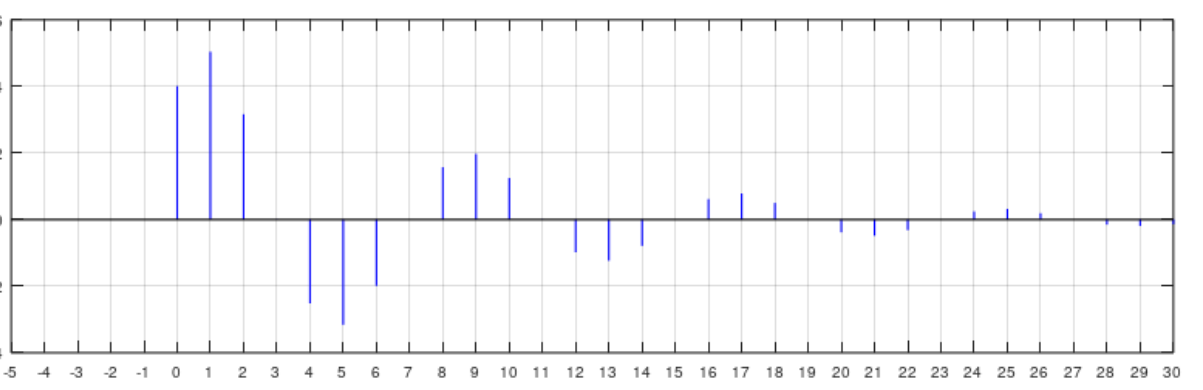

```
>> Xz=(6*z^2)/(z^2-sqrt(3)*z+1);
>> fn=itransformadaZ_Hz(Xz,1,false)
fn =

@(n) (+(2 * (6.00000) * ((0.50000) * cos (n * (0.52360)) +
 (0.86603) * sin (n * (0.52360))) .* f_u (n)))

>> n=[-5:30];
>> stem(n,fn(n),'b','marker','none');
```

$$7)\quad X(z)=\frac{4z^2}{z^2-\frac{8}{9}\sqrt{2}z+\frac{64}{81}}=4\left(\frac{z^2-\left(\frac{8}{9}\right)\left(\frac{\sqrt{2}}{2}\right)z+\left(\frac{8}{9}\right)\left(\frac{\sqrt{2}}{2}\right)z}{z^2-2\left(\frac{8}{9}\right)\left(\frac{\sqrt{2}}{2}\right)z+\left(\frac{8}{9}\right)^2}\right)=4\left(\frac{z^2-\left(\frac{8}{9}\right)\left(\frac{\sqrt{2}}{2}\right)z}{z^2-2\left(\frac{8}{9}\right)\left(\frac{\sqrt{2}}{2}\right)z+\left(\frac{8}{9}\right)^2}\right)+4\left(\frac{\left(\frac{8}{9}\right)\left(\frac{\sqrt{2}}{2}\right)z}{z^2-2\left(\frac{8}{9}\right)\left(\frac{\sqrt{2}}{2}\right)z+\left(\frac{8}{9}\right)^2}\right)$$

$$x[n]=4\sqrt{2}\left(\frac{8}{9}\right)^{\mathrm{n}}\left(\frac{\sqrt{2}}{2}\cos\left(\frac{\pi}{4}n\right)+\frac{\sqrt{2}}{2}\sin\left(\frac{\pi}{4}n\right)\right)\quad u[n];\quad |z|>3$$

$a^n\cos(bn)u(n)$	$\dfrac{z[z-a\cos(b)]}{z^2-[2a\cos(b)]z+a^2}$	$\lvert z\rvert>\lvert a\rvert$
$a^n\sin(bn)u(n)$	$\dfrac{[a\sin(b)]z}{z^2-[2a\cos(b)]z+a^2}$	$\lvert z\rvert>\lvert a\rvert$

$$a=\frac{8}{9}$$
$$\cos\left(\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}$$
$$\sin\left(\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}$$



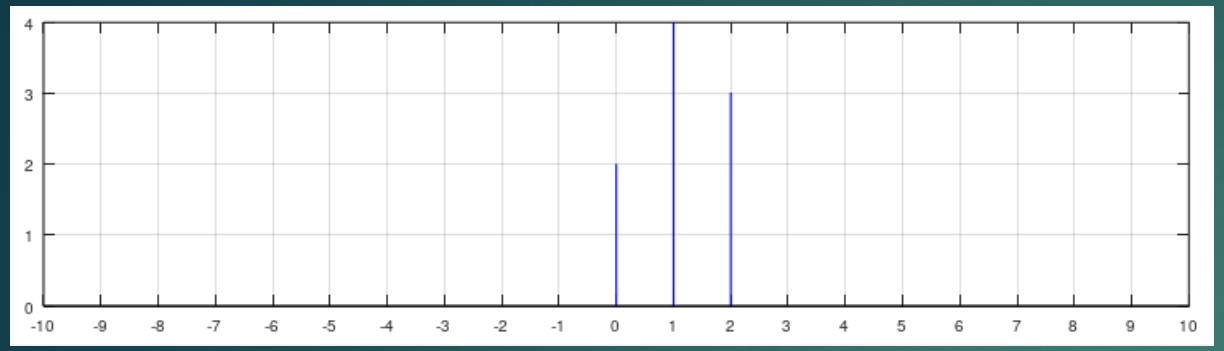

```
>> Xz=(4*z^2)/(z^2-8/9*sqrt(2)*z+64/81);
>> fn=itransformadaZ_Hz(Xz,1,false)
fn =

@(n) (+(2 * (2.82843) * (0.88889) .^ n .* ((0.70711) * cos (n * (0.78540))
 + (0.70711) * sin (n * (0.78540))) .* f_u (n)))

>> n=[-5:30];
>> stem(n,fn(n),'b','marker','none');
```

8) $X(z) = \quad 2 + 4z^{-1} + 3z^{-2}$

$\delta[n]$	1	$\forall z$

Observar que o polinômio em z, X(z), só tem numerador.
Por consequência, o sinal no tempo terá duração finita.

$$x[n] = 2\delta[n] + 4\delta[n-1] + \ 3\delta[n-2]$$



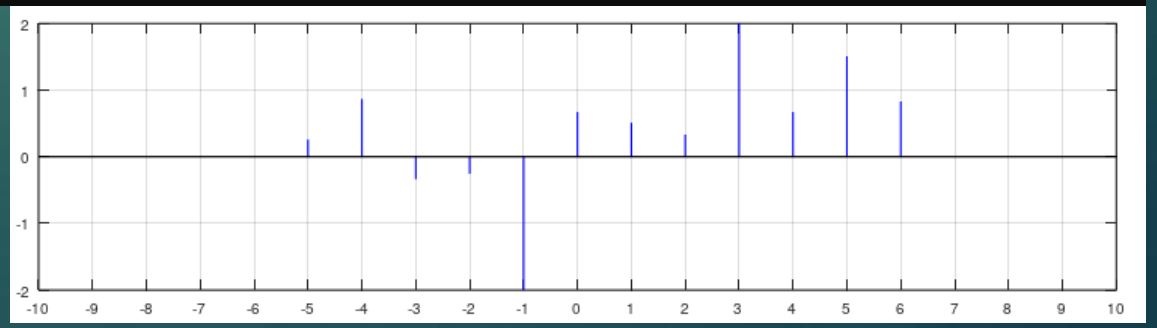

```
>> Xz = 2 + 4*z^-1 + 3*z^-2;
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+((2.00) * f_d (n)) + ((4.00) * f_d (n - 1)) + ((3.00) * f_d (n - 2)))

>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none')
```

9) $X(z) = \ \frac{1}{4}z^5 + \frac{6}{7}z^4 - \frac{1}{3}z^3 - \frac{1}{4}z^2 - 2z^1 + \frac{2}{3} + \frac{1}{2}z^{-1} + \frac{1}{3}z^{-2} + 2z^{-3} + \frac{2}{3}z^{-4} + \frac{3}{2}z^{-5} + \frac{5}{6}z^{-6}$

$\delta[n]$	1	$\forall z$

$$x[n] = \frac{1}{4}\delta[n+5] + \frac{6}{7}\delta[n+4] - \frac{1}{3}\delta[n+3] - \frac{1}{4}\delta[n+2] - 2\delta[n+1] + \frac{2}{3}\delta[n]\frac{1}{2}\delta[n-1] + \frac{1}{3}\delta[n-2] + 2\delta[n-3] + \frac{2}{3}\delta[n-4] + \frac{3}{2}\delta[n-5] + \frac{5}{6}\delta[n-6]$$

```
>> Xz = 1/4*z^5 +6/7*z^4 -1/3*z^3 -1/4*z^2 -2*z +2/3 +1/2*z^-1 +1/3*z^-2 +2*z^-3 +2/3*z^-4 +3/2*z^-5 +5/6*z^-6;
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+((0.25) * f_d (n + 5)) + ((0.86) * f_d (n + 4)) + ((-0.33) * f_d (n + 3)) + ((-0.25) * f_d (n + 2)) + ((-2.00) * f_d (n + 1)) + ((0.67) *
 f_d (n)) + ((0.50) * f_d (n - 1)) + ((0.33) * f_d (n - 2)) + ((2.00) * f_d (n - 3)) + ((0.67) * f_d (n - 4)) + ((1.50) * f_d (n - 5)) + ((0.83)
 * f_d (n - 6)))

>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none')
```



Expansão em frações Parciais

- Vemos pelas tabela de Transformada Z, que as funções X(z) apresentam, na sua maioria, um variável z livre, multiplicada ao numerador.
- Quando fazemos a expansão em frações parciais de X(z), obtemos um polinômio na forma $X(z) = \frac{A_1}{z-p_1} + \frac{A_2}{z-p_2} + \cdots + \frac{A_n}{z-p_n}$, onde $p_1, p_2, \ldots p_n$ são os polos de X(z) e $A_1, A_2, \ldots A_n$ os resíduos dos polos de X(z). No entanto, estas frações não têm a variável z livre no numerador!
- Para contornar este problema fazemos sempre a expansão de $\frac{X(z)}{z} = \frac{B_1}{z-p_1} + \frac{B_2}{z-p_2} + \cdots + \frac{B_n}{z-p_n}$, onde $B_1, B_2, \ldots B_n$ são os resíduos dos polos de $\frac{X(z)}{z}$. No final da expansão passamos denominador z multiplicando para o outro lado da igualdade, o que facilita a inspeção na tabela $X(z) = \frac{B_1 z}{z-p_1} + \frac{B_2 z}{z-p_2} + \cdots + \frac{B_n z}{z-p_n}$.

$x(n)$	$X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$	ROC
$\delta(n)$	1	$\lvert z\rvert > 0$
$au(n)$	$\frac{az}{z-1}$	$\lvert z\rvert > 1$
$nu(n)$	$\frac{z}{(z-1)^2}$	$\lvert z\rvert > 1$
$n^2 u(n)$	$\frac{z(z+1)}{(z-1)^3}$	$\lvert z\rvert > 1$
$a^n u(n)$	$\frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$
$e^{-na} u(n)$	$\frac{z}{(z-e^{-a})}$	$\lvert z\rvert > e^{-a}$
$na^n u(n)$	$\frac{az}{(z-a)^2}$	$\lvert z\rvert > \lvert a\rvert$
$\sin(an)u(n)$	$\frac{z\sin(a)}{z^2 - 2z\cos(a) + 1}$	$\lvert z\rvert > 1$
$\cos(an)u(n)$	$\frac{z[z - \cos(a)]}{z^2 - 2z\cos(a) + 1}$	$\lvert z\rvert > 1$
$a^n \sin(bn)u(n)$	$\frac{[a\sin(b)]z}{z^2 - [2a\cos(b)]z + a^2}$	$\lvert z\rvert > \lvert a\rvert$
$a^n \cos(bn)u(n)$	$\frac{z[z - a\cos(b)]}{z^2 - [2a\cos(b)]z + a^2}$	$\lvert z\rvert > \lvert a\rvert$
$e^{-an} \sin(bn)u(n)$	$\frac{[e^{-a}\sin(b)]z}{z^2 - [2e^{-a}\cos(b)]z + e^{-2a}}$	$\lvert z\rvert > e^{-a}$
$e^{-an} \cos(bn)u(n)$	$\frac{z[z - e^{-a}\cos(b)]}{z^2 - [2e^{-a}\cos(b)]z + e^{-2a}}$	$\lvert z\rvert > e^{-a}$
$2\lvert A\rvert\lvert P\rvert^n \cos(n\theta + \varphi)u(n)$ $P = \lvert P\rvert\angle\theta,\ A = \lvert A\rvert\angle\varphi$	$\frac{Az}{z-P} + \frac{A^* z}{z-P^*}$	

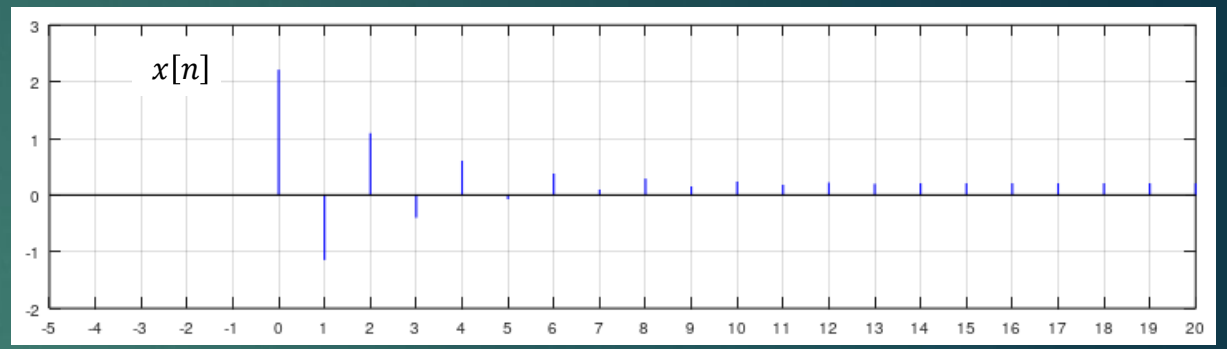
1) Polos Simples com Z no numerador

$$X(z) = \frac{\frac{11}{5} - \frac{28}{15}z^{-1}}{1 - \frac{1}{3}z^{-1} - \frac{2}{3}z^{-2}} = \frac{\frac{11}{5}z^2 - \frac{28}{15}z}{z^2 - \frac{1}{3}z - \frac{2}{3}}$$

$$\frac{X(z)}{z} = \frac{\frac{11}{5}z - \frac{28}{15}}{(z-1)\left(z+\frac{2}{3}\right)} = \frac{A}{z-1} + \frac{B}{z+\frac{2}{3}} = \frac{\frac{1}{5}}{z-1} + \frac{2}{z+\frac{2}{3}}$$

$$\text{então}: \quad X(z) = \frac{\frac{1}{5}z}{z-1} + \frac{2z}{z+\frac{2}{3}} \xrightarrow{z^{-1}} \frac{1}{5}u[n] + 2\left(-\frac{2}{3}\right)^n u[n]$$

```
>> Xz = (11/5*z^2-28/15*z)/(z^2-1/3*z-2/3);
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+(0.20000 * f_u (n)) + (2.00000 * (-0.66667) .^ n .* f_u (n)))

>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none')
```



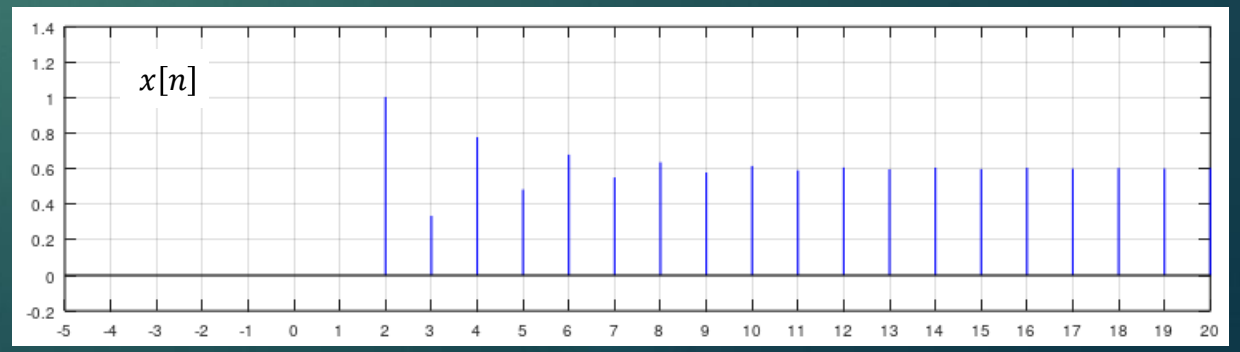
2) Polos Simples sem z no numerador

$$X(z) = \frac{z^{-2}}{1 - \frac{1}{3}z^{-1} - \frac{2}{3}z^{-2}} = \frac{1}{z^2 - \frac{1}{3}z - \frac{2}{3}} = \frac{1}{(z-1)\left(z+\frac{2}{3}\right)}$$

$$\frac{X(z)}{z} = \frac{1}{z(z-1)\left(z+\frac{2}{3}\right)} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{z+\frac{2}{3}} = \frac{-\frac{3}{2}}{z} + \frac{\frac{3}{5}}{z-1} + \frac{\frac{9}{10}}{z+\frac{2}{3}}$$

$$\text{então}: \quad X(z) = -\frac{3}{2} + \frac{\frac{3}{5}z}{z-1} + \frac{\frac{9}{10}z}{z+\frac{2}{3}} \xrightarrow{z^{-1}} -\frac{3}{2}\delta[n] + \frac{3}{5}u[n] + \frac{9}{10}\left(-\frac{2}{3}\right)^n u[n]$$

```
>> Xz = 1/((z-1)*(z+2/3));
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+(0.60000 * f_u (n)) + (0.90000 * (-0.66667) .^ n .* f_u (n)) +
 (-1.50000 * f_d (n)))

>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none')
```



3) Polos simples com Fração Imprópria

$$2z^3 - \frac{2}{3}z^2 + \frac{13}{15}z - \frac{28}{15} \quad \Big| \; z^2 - \frac{1}{3}z - \frac{2}{3}$$

$$2z^3 - \frac{2}{3}z^2 + \frac{4}{3}z \qquad\qquad 2z$$

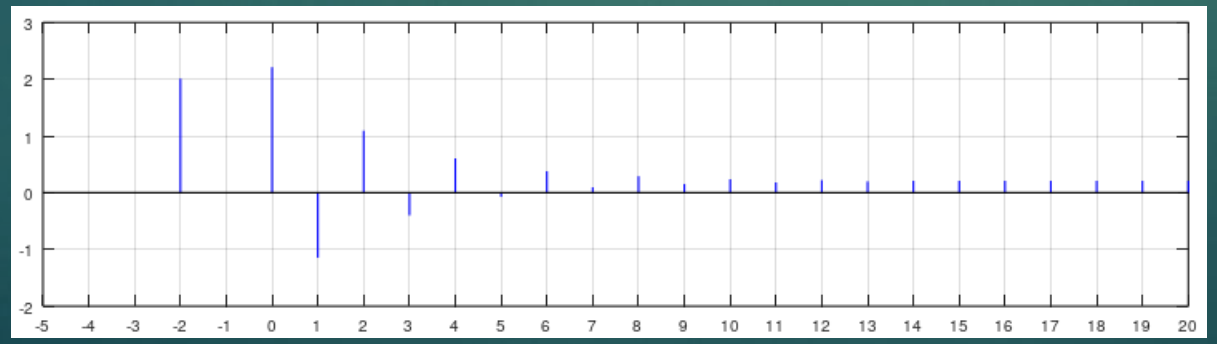
$$\frac{11}{5}z - \frac{28}{15}$$

$$X(z) = \frac{2 - \frac{2}{3}z^{-1} + \frac{13}{15}z^{-2} - \frac{28}{15}z^{-3}}{z^{-2} - \frac{1}{3}z^{-3} - \frac{2}{3}z^{-4}} = \frac{2z^4 - \frac{2}{3}z^3 + \frac{13}{15}z^2 - \frac{28}{15}z}{z^2 - \frac{1}{3}z - \frac{2}{3}}$$

$$\frac{X(z)}{z} = \frac{2z^3 - \frac{2}{3}z^2 + \frac{13}{15}z - \frac{28}{15}}{z^2 - \frac{1}{3}z - \frac{2}{3}} = 2z + \frac{\frac{11}{5}z - \frac{28}{15}}{(z-1)\left(z + \frac{2}{3}\right)} = 2z + \frac{A}{z-1} + \frac{B}{z + \frac{2}{3}} = 2z + \frac{\frac{1}{5}}{z-1} + \frac{2}{z + \frac{2}{3}}$$

$$X(z) = 2z^2 + \frac{\frac{1}{5}z}{z-1} + \frac{2z}{z + \frac{2}{3}} \xrightarrow{z^{-1}} 2\delta[n+2] + \frac{1}{5}u[n] + 2\left(-\frac{2}{3}\right)^n u[n]$$

```
>> Xz = (2*z^4-2/3*z^3+13/15*z^2-28/15*z)/(z^2-1/3*z-2/3);
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+((2.00) * f_d (n + 2)) + (0.20000 * f_u (n)) + (2.00000
 * (-0.66667) .^ n .* f_u (n)))

>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none')
```



4) Polo duplo

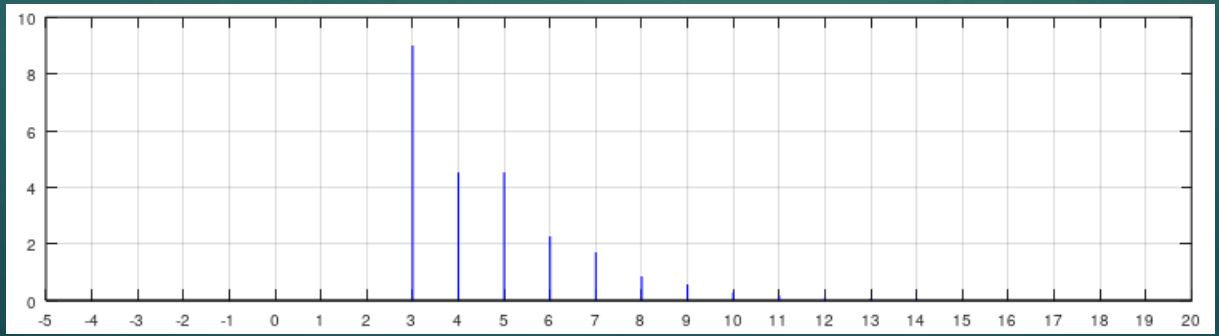
$na^n u(n)$	$\frac{az}{(z-a)^2}$	$\lvert z\rvert > \lvert a\rvert$

$$X(z) = \frac{9z^{-3}}{\left(1+\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{2}z^{-1}\right)^2} = \frac{9}{\left(z+\frac{1}{2}\right)\left(z-\frac{1}{2}\right)^2}$$

$$\frac{X(z)}{z} = \frac{9}{(z)\left(z+\frac{1}{2}\right)\left(z-\frac{1}{2}\right)^2} = \frac{\mathrm{A}}{\mathrm{z}} + \frac{\mathrm{B}}{\mathrm{z}+\frac{1}{2}} + \frac{C}{(\mathrm{z}-\frac{1}{2})} + \frac{D}{\left(\mathrm{z}-\frac{1}{2}\right)^2} = \frac{72}{\mathrm{z}} + \frac{-18}{\mathrm{z}+\frac{1}{2}} + \frac{-54}{(\mathrm{z}-\frac{1}{2})} + \frac{18}{\left(\mathrm{z}-\frac{1}{2}\right)^2}$$

$$X(z) = 72 + \frac{-18z}{\mathrm{z}+\frac{1}{2}} + \frac{-54z}{(\mathrm{z}-\frac{1}{2})} + \frac{36(\frac{1}{2})z}{\left(\mathrm{z}-\frac{1}{2}\right)^2} \xrightarrow{Z^{-1}} 72\delta[n] - 18\left(-\frac{1}{2}\right)^n u[n] - 54\left(\frac{1}{2}\right)^n u[n] + 36\mathrm{n}\left(\frac{1}{2}\right)^{\mathrm{n}} u[n]$$

```
>> Xz=9/((z+1/2)*(z-1/2)^2);
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+(-54.00000 * (0.50000) .^ n .* f_u (n)) + (36.00000 * n .* (0.50000) .^ n .* f_u (n)) +
 (-18.00000 * (-0.50000) .^ n .* f_u (n)) + (72.00000 * f_d (n)))

>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none')
```



5) Polos Complexos Conjugados

$$2|A||P|^n\cos(n\theta+\varphi)u(n)$$
$$P=|P|\angle\theta,\ A=|A|\angle\varphi$$

$$\frac{Az}{z-P}+\frac{A^*z}{z-P^*}$$

$$X(z)=\frac{\left(\frac{1}{2}-\frac{1}{12}z^{-1}\right)}{\left(1+\frac{1}{3}z^{-1}+\frac{5}{36}z^{-2}\right)}=\frac{\left(\frac{1}{2}z^2-\frac{1}{12}z\right)}{\left(z^2+\frac{1}{3}z+\frac{5}{36}\right)}$$

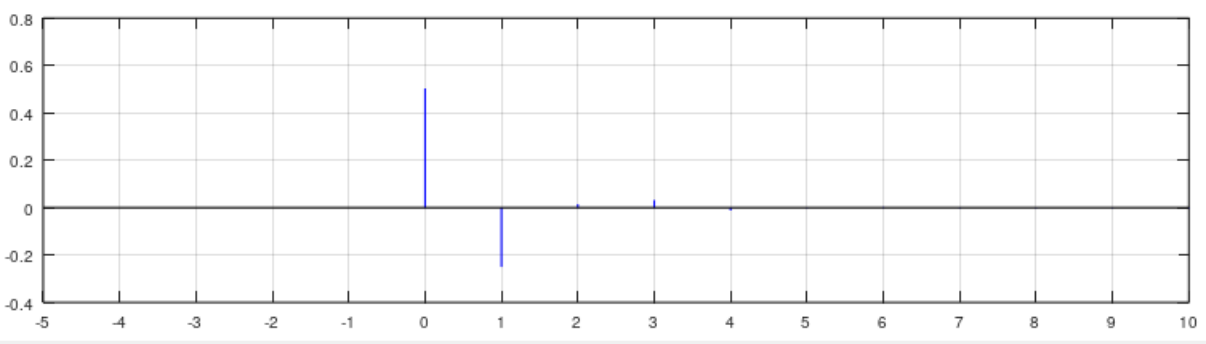
$$A=\frac{1}{4}+j\frac{1}{4}=\frac{\sqrt{2}}{4}e^{jatan\left(\frac{1/4}{1/4}\right)}=\frac{\sqrt{2}}{4}e^{\frac{j\pi}{4}}=\frac{\sqrt{2}}{4}e^{j\,0.7853982}$$

$$P=-\frac{1}{6}+j\frac{1}{3}=\frac{\sqrt{5}}{6}e^{\mathrm{jatan}\left(\frac{1/3}{-1/6}\right)}=\frac{\sqrt{5}}{6}e^{j2.0344559}$$

$$\frac{X(z)}{z}=\frac{\left(\frac{1}{2}z-\frac{1}{12}\right)}{\left(z^2+\frac{1}{3}z+\frac{5}{36}\right)}=\frac{\mathrm{A}}{\mathrm{z}+\frac{1}{6}-\mathrm{j}\frac{1}{3}}+\frac{\mathrm{A}^*}{\mathrm{z}+\frac{1}{6}+\mathrm{j}\frac{1}{3}}=\frac{\frac{1}{4}+\mathrm{j}\frac{1}{4}}{\mathrm{z}+\frac{1}{6}-\mathrm{j}\frac{1}{3}}+\frac{\frac{1}{4}-\mathrm{j}\frac{1}{4}}{\mathrm{z}+\frac{1}{6}+\mathrm{j}\frac{1}{3}}$$

$$X(z)=\frac{\left(\frac{1}{4}+\mathrm{j}\frac{1}{4}\right)z}{\mathrm{z}+\frac{1}{6}-\mathrm{j}\frac{1}{3}}+\frac{\left(\frac{1}{4}-\mathrm{j}\frac{1}{4}\right)z}{\mathrm{z}+\frac{1}{6}+\mathrm{j}\frac{1}{3}}\xrightarrow{z^{-1}}2\left(\frac{\sqrt{2}}{4}\right)\left(\frac{\sqrt{5}}{6}\right)^n\cos(n2.0344559+0.7853982)u[n]$$

```
>> Xz=(1/2*z^2-1/12*z)/(z^2+1/3*z+5/36);
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+(2 * (0.35355) * (0.37268) .^ n .* cos (n * (2.03444) - (-0.78540)) .* f_u (n)))

>> n=[-5:10];
>> stem(n,fn(n),'b','marker','none')
```



Divisão Longa

$$X(z) = \frac{2z^2}{z^2 - 2z + 1}$$

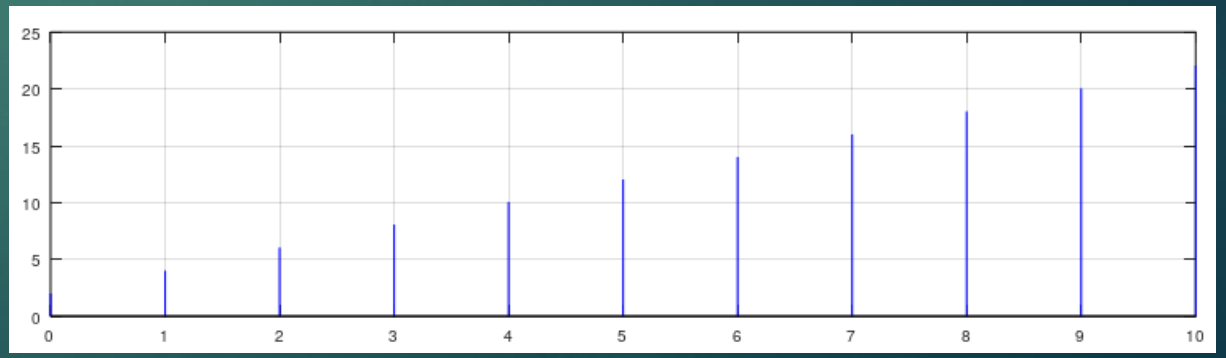
$$X(z) = \frac{2z^2}{z^2-2z+1} = 2 + 4z^{-1} + 6z^{-2} + 8z^{-3} + 10z^{-4} + \ldots..$$

$$x[n] = 2\delta[n] + 4\delta[n-1] + 6\delta[n-2] + 8\delta[n-3] + 10\delta[n-4] + \ldots..$$

$$x[n] = [2, 4, 6, 8, 10, \ldots]$$

$$\begin{array}{l|l}
2z^2 & z^2 - 2z + 1 \\
2z^2 - 4z + 2 & \overline{} \\
\hline
0 + 4z - 2 & 2 + 4z^{-1} + 6z^{-2} + 8z^{-3} + 10z^{-4} + \ldots.. \\
\quad 4z - 8 + 4z^{-1} & \\
\hline
\quad 0 + 6 - 4z^{-1} & \\
\qquad 6 - 12z^{-1} + 6z^{-2} & \\
\hline
\qquad 0 + 8z^{-1} - 6z^{-2} & \\
\qquad\quad 8z^{-1} - 16z^{-2} + 8z^{-3} & \\
\hline
\qquad\quad 0 + 10z^{-2} - 6z^{-3} & \ldots..
\end{array}$$

```
>> Xz=(2*z^2)/(z^2-2*z+1);
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+(2.00000 * f_u (n)) + (2.00000 .* n .* f_u (n)))

>> n=[0:10];
>> stem(n,fn(n),'b','marker','none')
>> fn(n)
ans =

    2    4    6    8   10   12   14   16   18   20   22
```



Integral de Contorno

- Vamos definir matematicamente a Transformada Z inversa. Começamos pela transformada direta X(z) do sinal no tempo discreto x(n);

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \qquad \text{Mas} \qquad z = re^{jw} \qquad X(z) = X(z = re^{jw}) = \sum_{n=-\infty}^{\infty} (r^{-n}x[n])\ e^{-jwn}$$

Então $X(z) = X(re^{jw})$ é a transformada de Fourier do tempo discreto de $(r^{-n}x[n])$!
Usando então a Transformada inversa de Fourier do Tempo Discreto de $X(z)$ teremos:

$$(r^{-n}x[n]) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(z)e^{jwn}\,dw \qquad \text{Como } z = re^{jw} \quad dz = jre^{jw}dw = jzdw \quad dw = \frac{dz}{jz}$$

$$(r^{-n}x[n]) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(z)e^{jwn}\frac{\mathrm{dz}}{\mathrm{jz}}$$

$$\mathrm{x}[n] = \frac{1}{2\pi j}\int_{-\infty}^{\infty} X(z)r^{n}e^{jwn}\frac{\mathrm{dz}}{\mathrm{z}} = \frac{1}{2\pi j}\int_{-\infty}^{\infty} X(z)z^{n-1}\,dz$$

$$\boxed{\mathrm{x}[n] = \frac{1}{2\pi j}\int_{-\infty}^{\infty} X(z)z^{n-1}\,dz = \sum_{k=1}^{K} R_k}$$

- $\mathrm{x}[n]$, que é a transformada Z inversa de X(z)
- Esta integral pode ser avaliada pelo Teorema de Cauchy, a partir da soma dos resíduos R_k dos k polos de $X(z)z^{n-1}$

1) Polos Simples

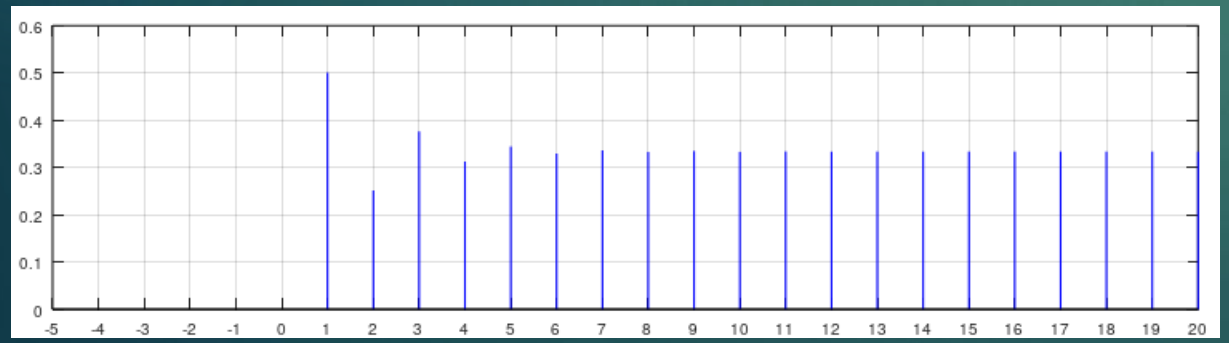
$$X(z) = \frac{1}{2}\frac{z^{-1}}{(1-z^{-1})\left(1+\frac{1}{2}z^{-1}\right)} = \frac{\frac{1}{2}z}{(z-1)\left(z+\frac{1}{2}\right)}$$

Pela integral de contorno:

$$R_1 = X(z)z^{n-1}\left(z+\frac{1}{2}\right)\Bigg|_{z=-\frac{1}{2}} \qquad R_1 = \frac{\frac{1}{2}z}{(z-1)\cancel{\left(z+\frac{1}{2}\right)}}z^{n-1}\cancel{\left(z+\frac{1}{2}\right)}\Bigg|_{z=-\frac{1}{2}} = \frac{\frac{1}{2}\left(-\frac{1}{2}\right)^n}{(-\frac{1}{2}-1)} = -\frac{1}{3}\left(-\frac{1}{2}\right)^n$$

$$R_2 = X(z)z^{n-1}(z-1)\Bigg|_{z=1} \qquad R_2 = \frac{\frac{1}{2}z}{\cancel{(z-1)}\left(z+\frac{1}{2}\right)}z^{n-1}\cancel{(z-1)}\Bigg|_{z=1} = \frac{\frac{1}{2}(1)^n}{(1+\frac{1}{2})} = \frac{1}{3}$$

$$\mathrm{x}[n] = \sum_{k=1}^{2} R_k = -\frac{1}{3}\left(-\frac{1}{2}\right)^n u[n] + \frac{1}{3}\mathrm{u}[n]$$




```
>> Xz=(1/2*z)/((z-1)*(z+1/2));
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+(0.33333 * f_u (n)) + (-0.33333 * (-0.50000) .^ n .* f_u (n)))

>> n=[-5:20];
>> stem(n,fn(n),'b','marker','none')
```

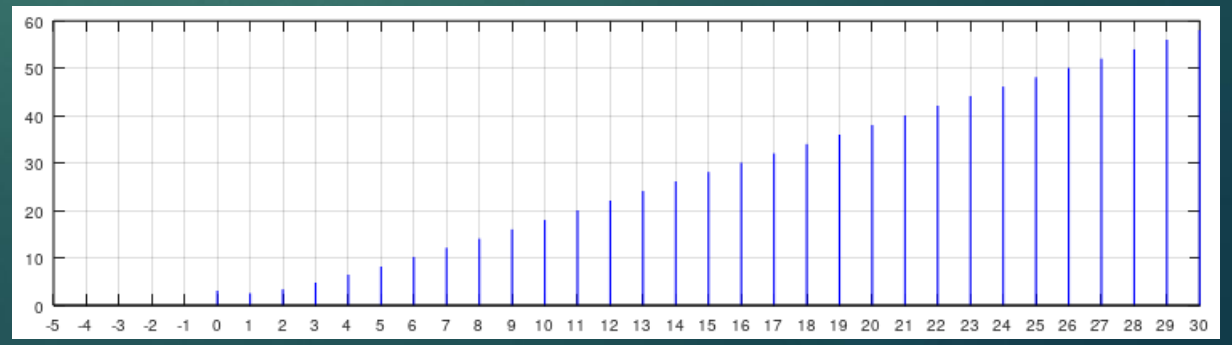
2) Polo Duplo + Polos Simples

$$X(z) = \frac{3z^3 - 5z^2 + 3z}{(z-1)^2\left(z - \frac{1}{2}\right)}$$

Pela integral de contorno:

$$R_1 = X(z)z^{n-1}\left(z - \frac{1}{2}\right)\Bigg|_{z=\frac{1}{2}} \qquad R_1 = \frac{3z^3 - 5z^2 + 3z}{(z-1)^2} z^{n-1}\Bigg|_{z=\frac{1}{2}} = \frac{3\left(\frac{1}{2}\right)^2 - 5\left(\frac{1}{2}\right) + 3}{\left(\frac{1}{2} - 1\right)^2}\left(\frac{1}{2}\right)^n = 5\left(\frac{1}{2}\right)^n$$

$$R_2 = \frac{1}{(p-1)!}\frac{d^{p-1}}{dz^{p-1}}\left[X(z)z^{n-1}(z-1)^2\right]\Bigg|_{z=1} \qquad R_2 = \frac{1}{1!}\frac{d}{dz}\left[\frac{3z^3 - 5z^2 + 3z}{\left(z - \frac{1}{2}\right)} z^{n-1}\right]\Bigg|_{z=1} = 2n - 2$$

$$\mathrm{x}[n] = \sum_{k=1}^{K} R_k = 5\left(\frac{1}{2}\right)^n \mathrm{u}[n] + (2n-2)u[n]$$

```
>> Xz=(3*z^3-5*z^2+3*z)/((z-1)^2*(z-1/2));
>> fn=itransformadaZ_Hz(Xz,1,true)
fn =

@(n) (+(-2.00000 * f_u (n)) + (2.00000 .* n .* f_u (n)) + (5.00000 * (0.50000) .^ n .* f_u (n)))

>> n=[-5:30];
>> stem(n,fn(n),'b','marker','none')
```